

Streaming Data Science Challenging the i.i.d. assumption

Emanuele Della Valle

Prof @ PoliMI

CRO & founder @ Motus ml

founder @ Quantia Consulting



POLITECNICO
MILANO 1863



POLITECNICO
MILANO 1863

The vision **Continuity matters!**

recall



POLITECNICO
MILANO 1863

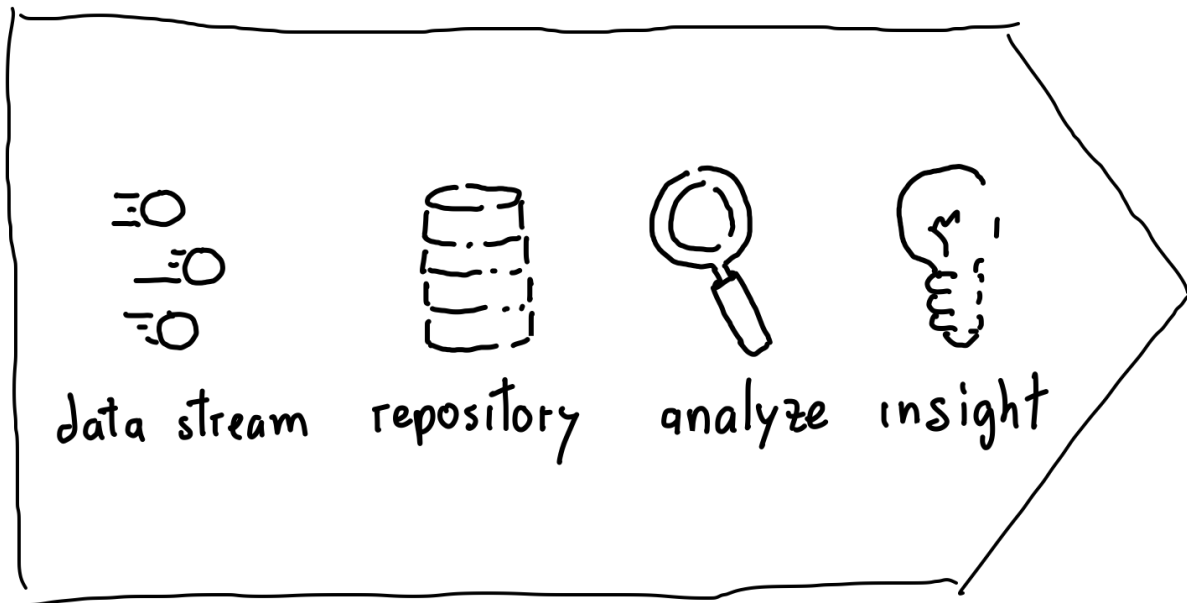
Continuity matters!



Photo by
recall

Traditional approach vs ...

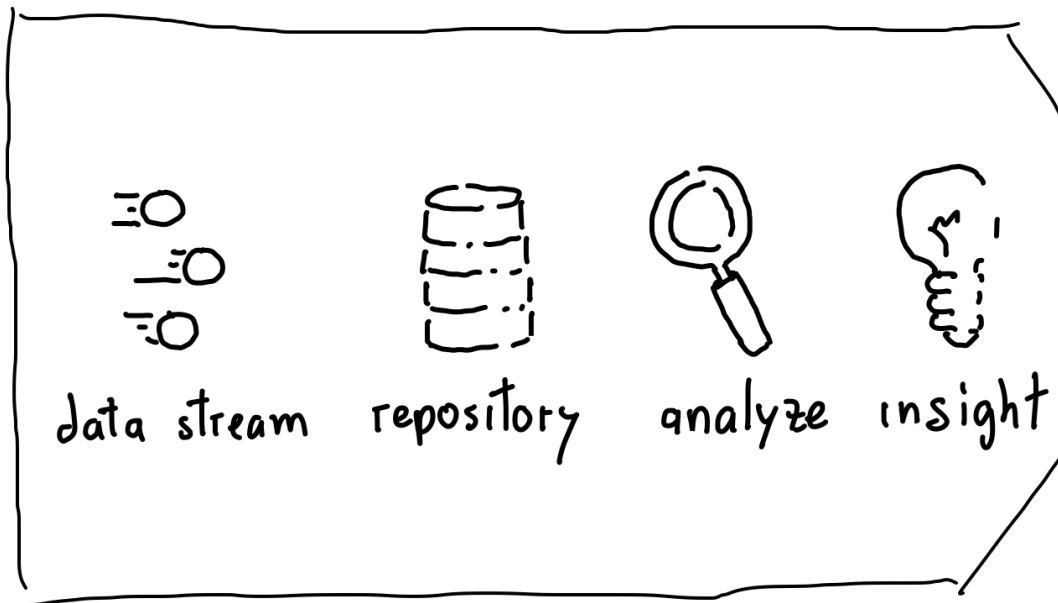
Traditional approach



Stop data to analyze

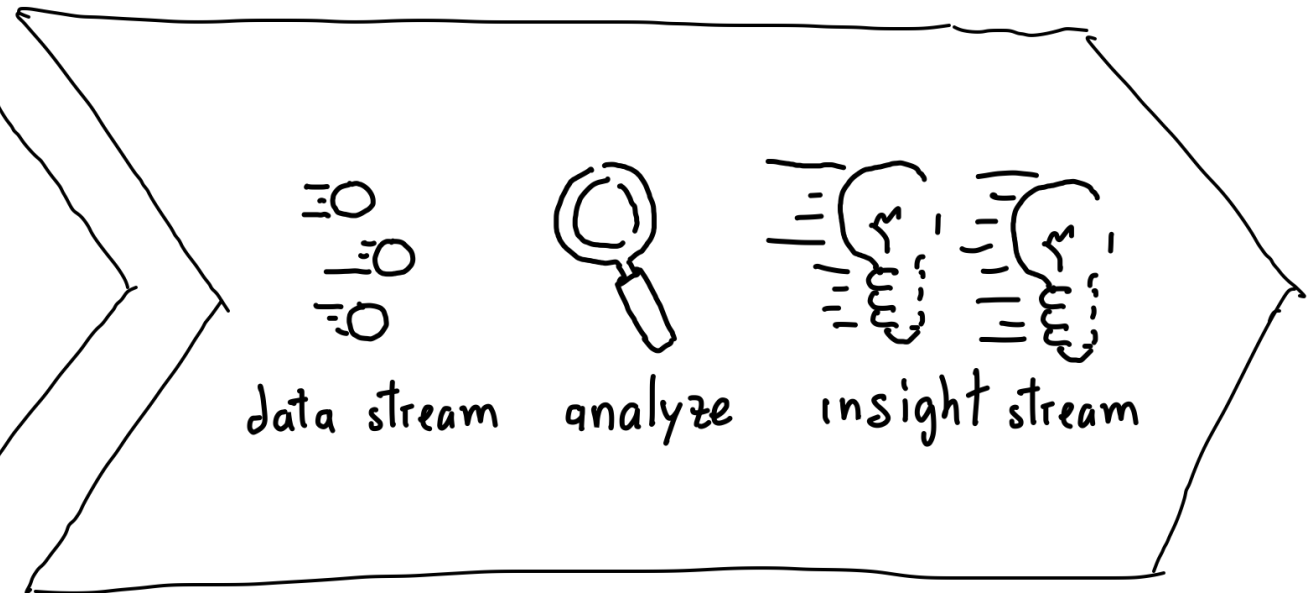
Traditional vs. Continuous approach

Traditional approach



Stop data to analyze

Continuous approach



Analyze data in motion

recall



The key ingredients to enable continuity

1. Sensor and actuators
2. Connectivity
3. Streaming Data Engineering
4. Streaming Data Science



The key ingredients to enable continuity

1. Sensor and actuators
2. Connectivity
- 3. Streaming Data Engineering**
4. Streaming Data Science



Event
Based
Systems



Data
Stream
Management
Systems



Complex
Event
Processing



Event
Driven
Architecture



The key ingredients to enable continuity

1. Sensor and actuators
2. Connectivity
3. Streaming Data Engineering
4. **Streaming Data Science**

ARTIFICIAL INTELLIGENCE

Our weird behavior during the pandemic is messing with AI models

Machine-learning models trained on normal behavior are showing cracks — forcing humans to step in to set them straight.

By Will Douglas Heaven

May 11, 2020



POLITECNICO
MILANO 1863

[src: <https://www.technologyreview.com/2020/05/11/1001563/covid-pandemic-broken-ai-machine-learning-amazon-retail-fraud-humans-in-the-loop/>]



Example of changes in natural language

- The word **Virus** in the **2010s** referred 55% of the time to **computers**, **suddenly** in **2020**, it referred 85% of the time to **medicine**
- **Mouse** in the **70s** referred 95% of the time to the **animal** slowly in the **90s** it started to refer **also to the device** and in the **2000s** it referred 85% of the time to the **device**
- **Turkey** in **July every year** refers 60% of the time to the **animal** in **November**, it refers 70% of the time to the **food**
- The word **selfie** before the **2010s** practically did not exist



Example of changes in computer vision

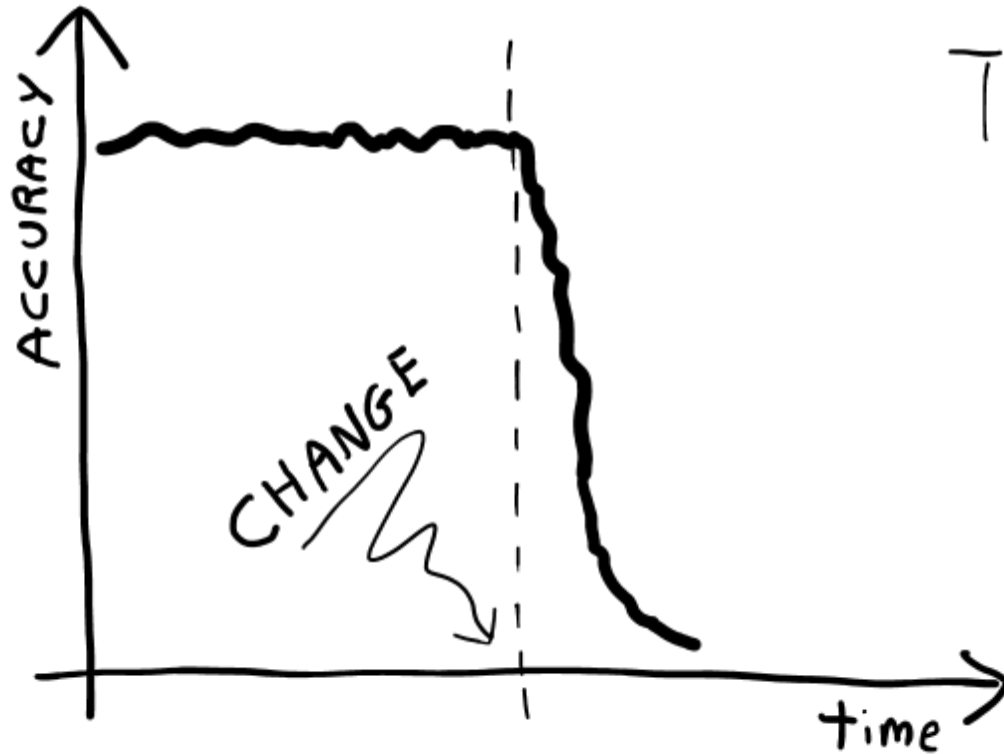
- **Sudden changes** in the **acquisition device**, e.g., an x-ray machine 100 years ago vs a modern one
- **Slow changes** in the **status of the device**, e.g., lenses getting dirty
- **Seasonal changes** in the **environment**, e.g., road signals on sunny summer days vs. snowy winter ones
- **Recurring changes** in the **object**, e.g., players wearing different uniforms in different matches at home or away
- **Changes** can introduce **new objects**, e.g., the appearance of smartphones



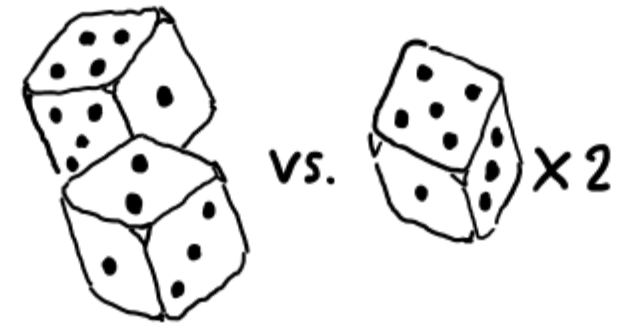
Other examples of changes

- Churn reasons for customers of services
- Movies to recommend on IPTVs
- Survival rate in medicine
- ...

Changes cause AI models to lose relevancy



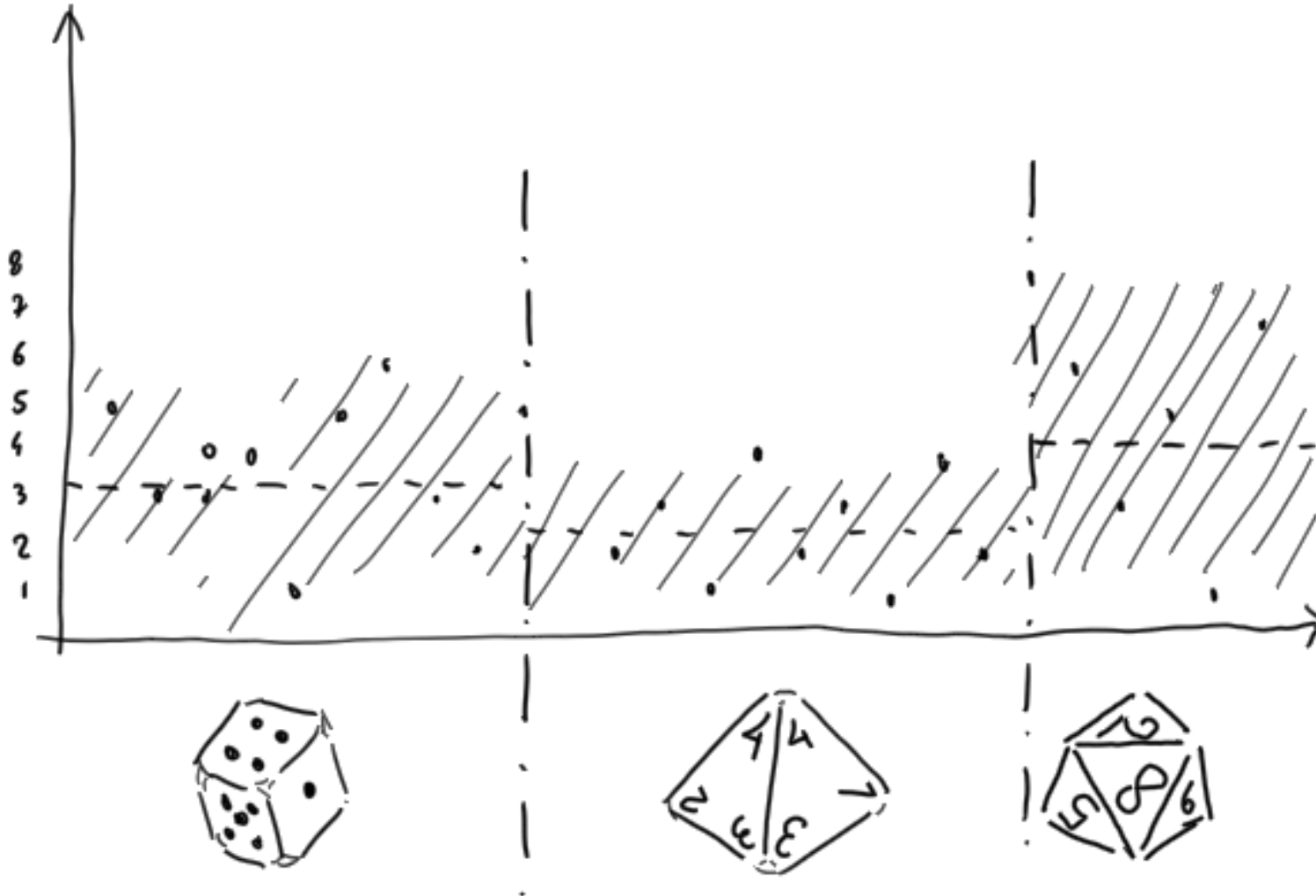
The assumption: data is
independent &
identically
distributed



It does **not** hold

i.i.d. assumption explained

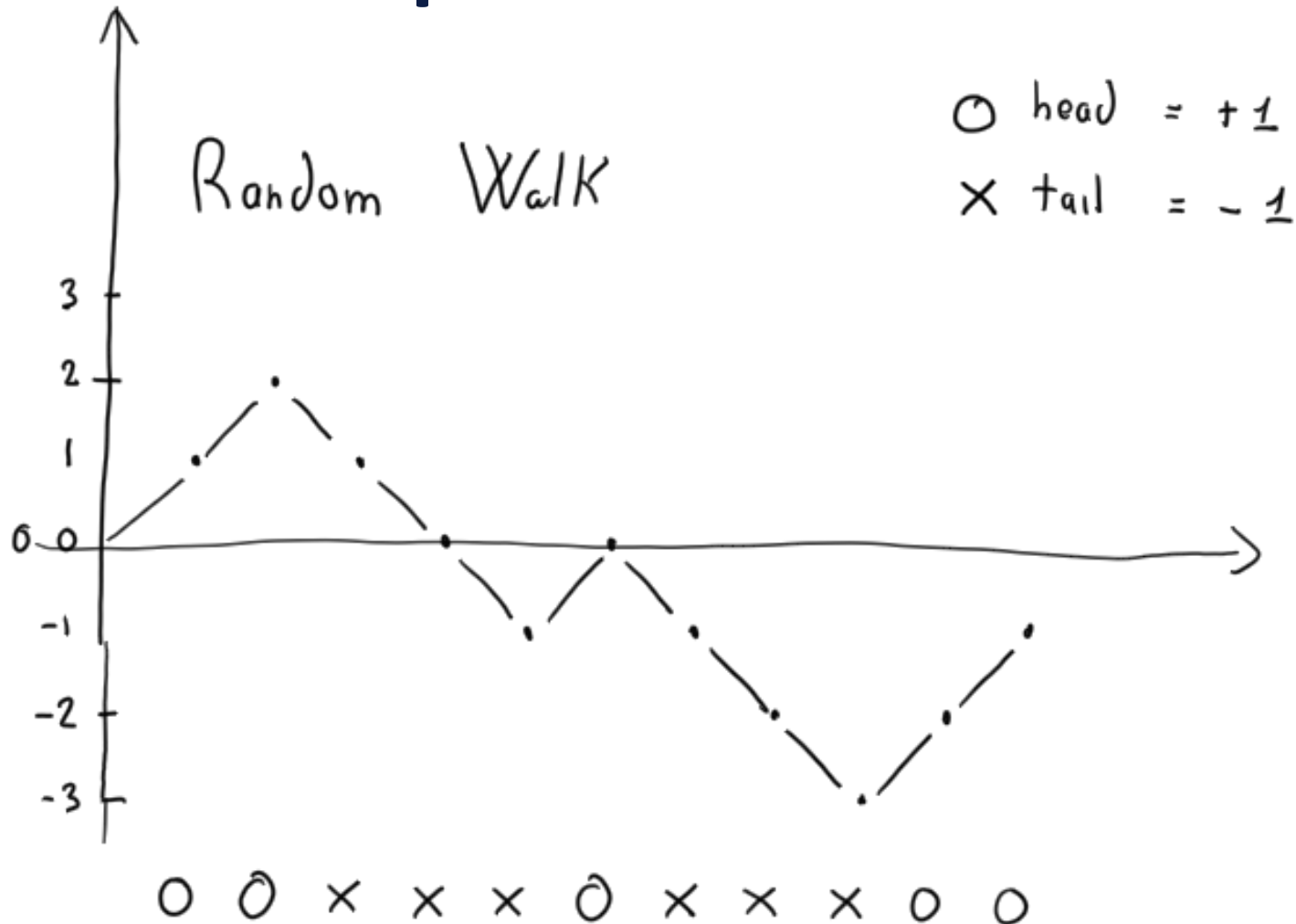
Non identically distributed data



CHANGE HAPPENS !!!

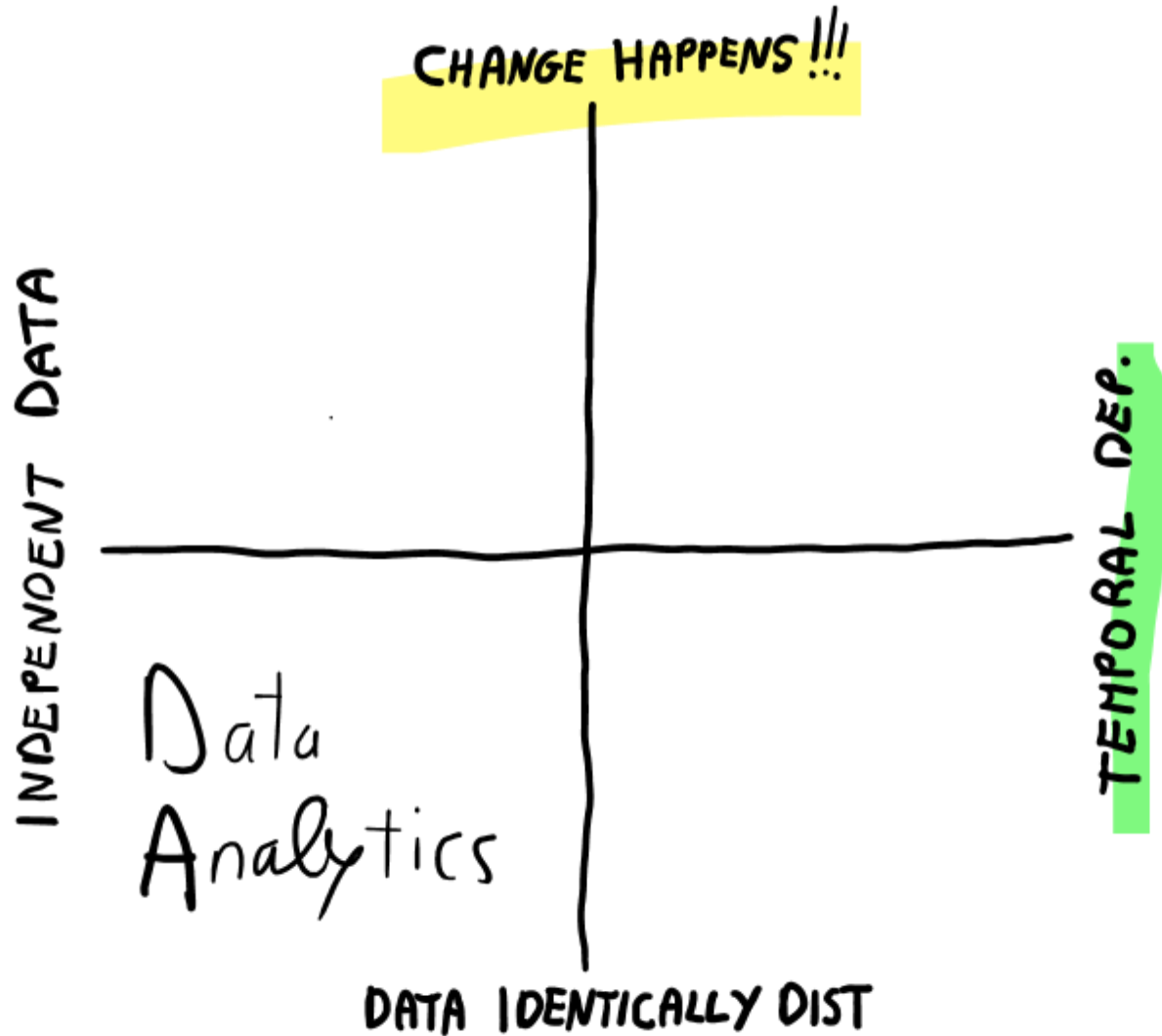
i.i.d. assumption explained

Non independent data

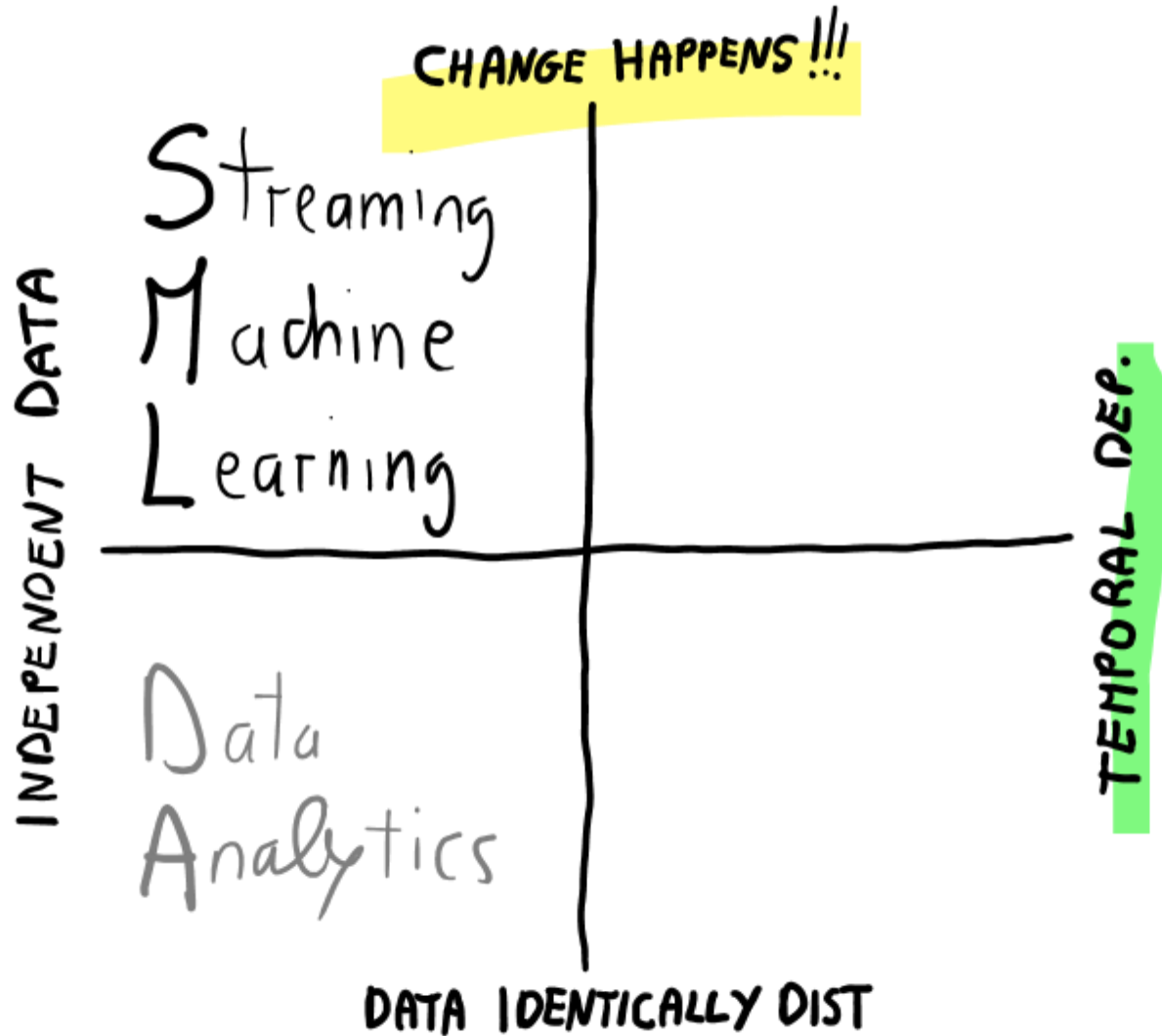


TEMPORAL DEP.

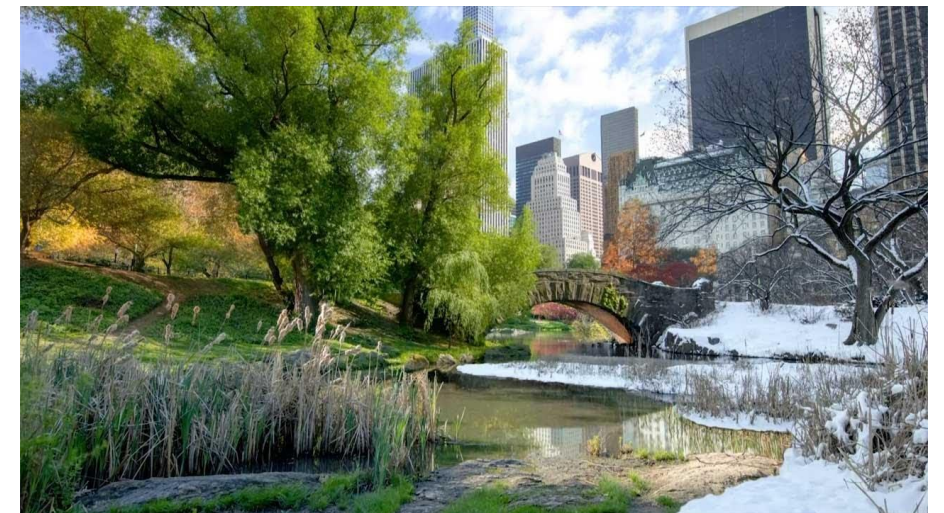
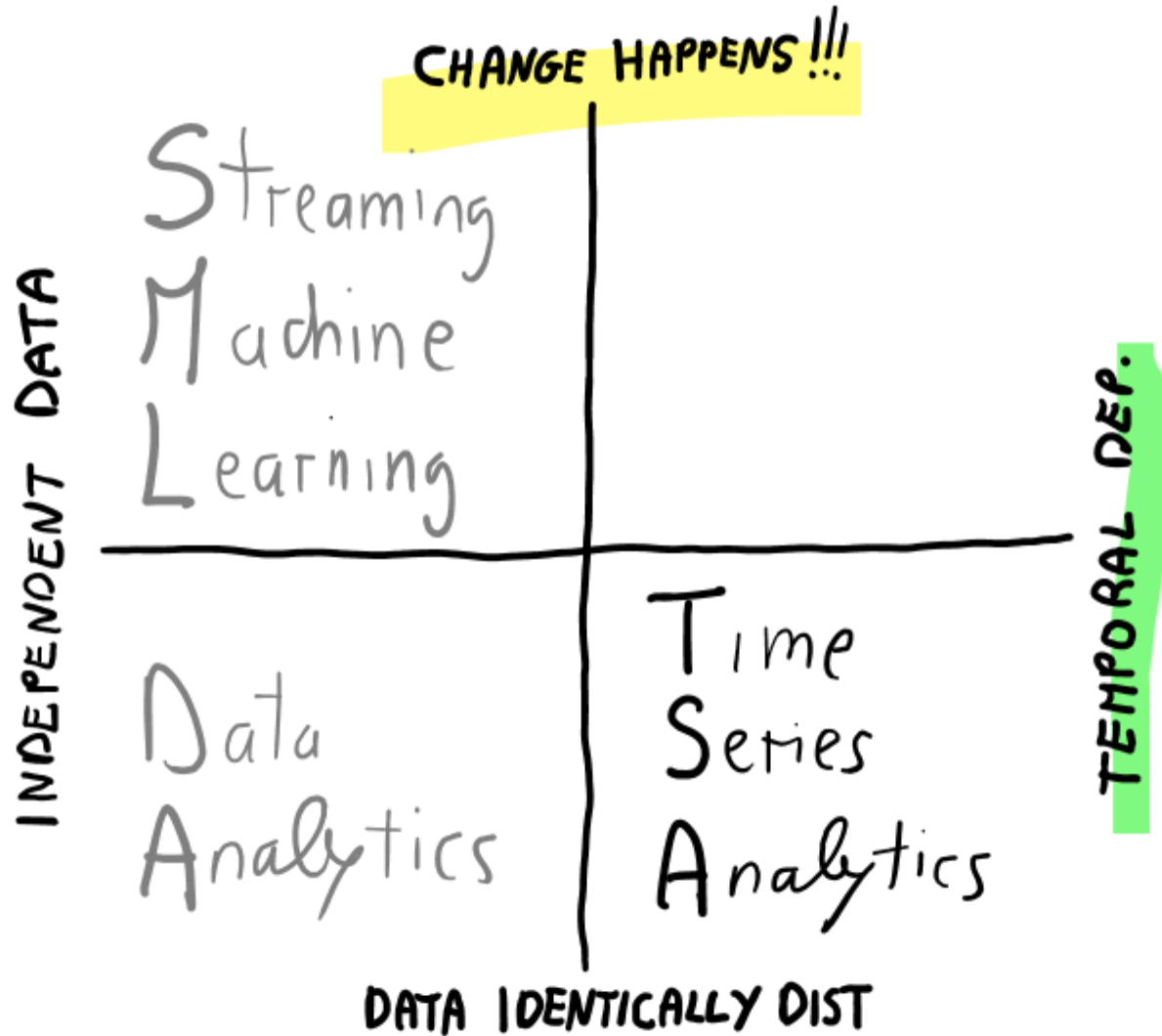
State-of-the-art



State-of-the-art



State-of-the-art



Explaining the past and forecast the future
of a continuous flow of data
without assuming data independence
with
Time Series Analytics



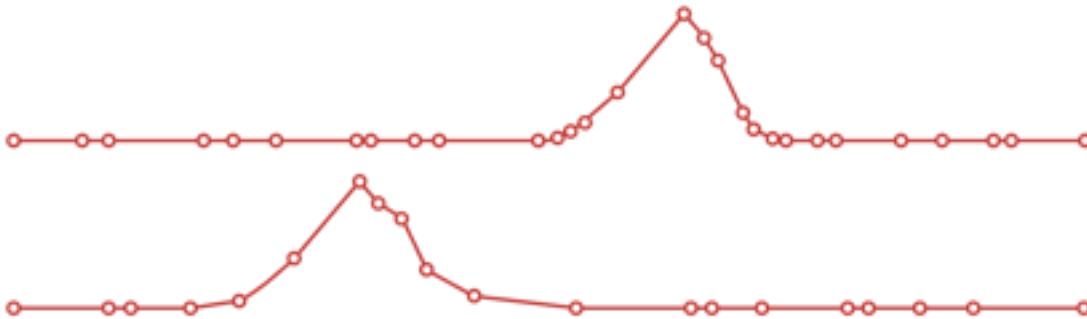
Type of data

A *time series* is a sequence of observations on **one** (or more) **quantitative** variable **regularly** collected **over time**.

Events vs time series

The phenomenon happens
and we observe them

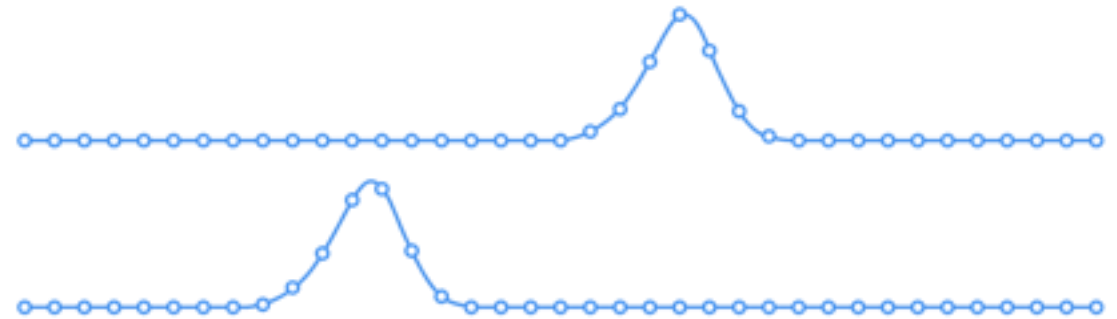
irregularly



Events

We monitor a
phenomenon

regularly



Time series

Events becomes time series via windowing

E.g.,

- The average trade price of Apple stock every 10 minutes over the course of a day
- The average response time for requests in an application over 1 minute intervals
- ...

Two purposes

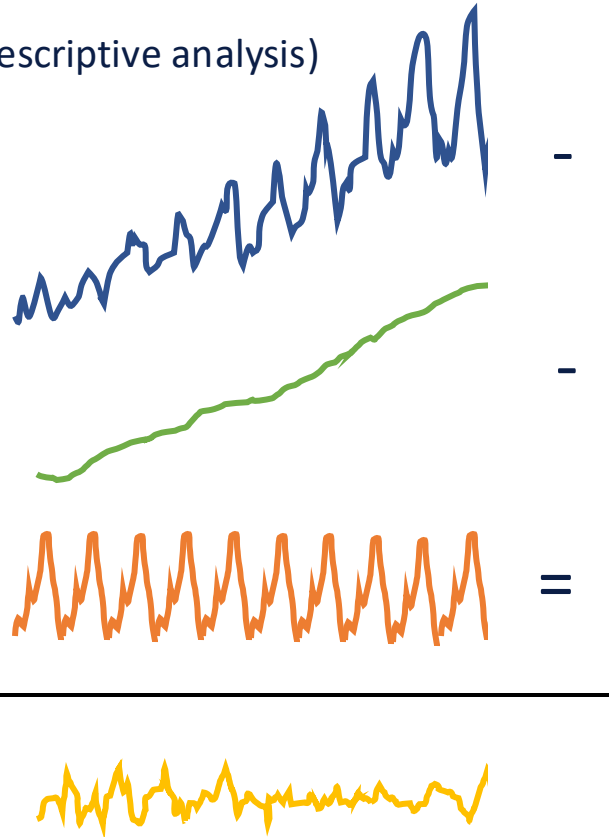
Modeling (a.k.a., descriptive analysis)

- Original

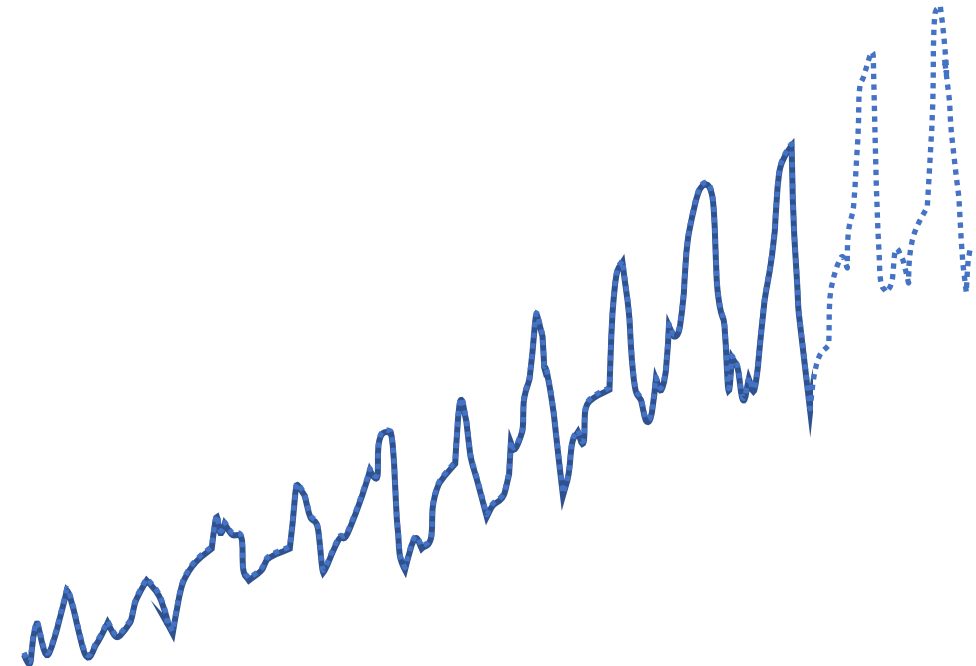
- Trend

- Seasonality

- Residual



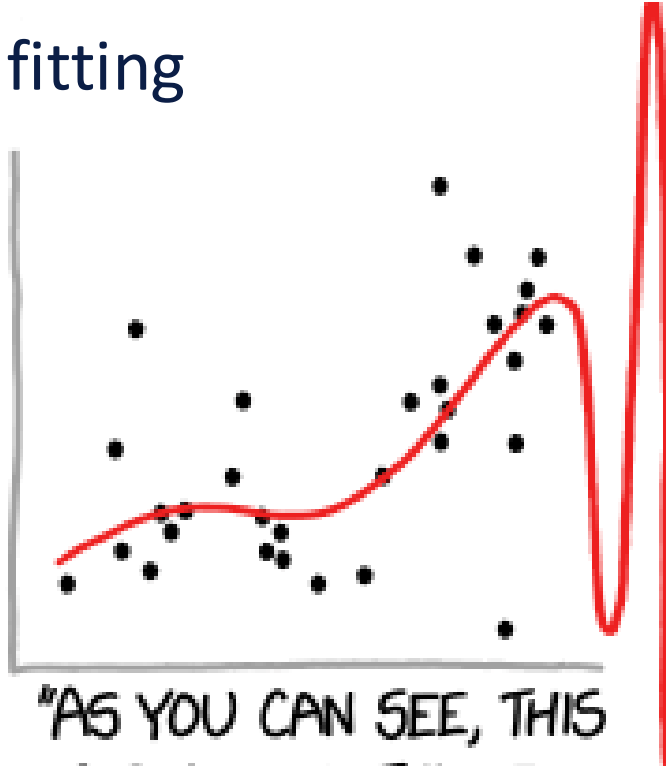
- **Forecasting** (a.k.a., predictive analysis)



Modeling traditional techniques

Capturing the Trend

Not fitting



[src: <https://xkcd.com/2048/>]

"AS YOU CAN SEE, THIS
MODEL SMOOTHLY FITS
THE— WAIT NO NO DON'T
EXTEND IT AAAAAA!!"

But **smoothing** or ...



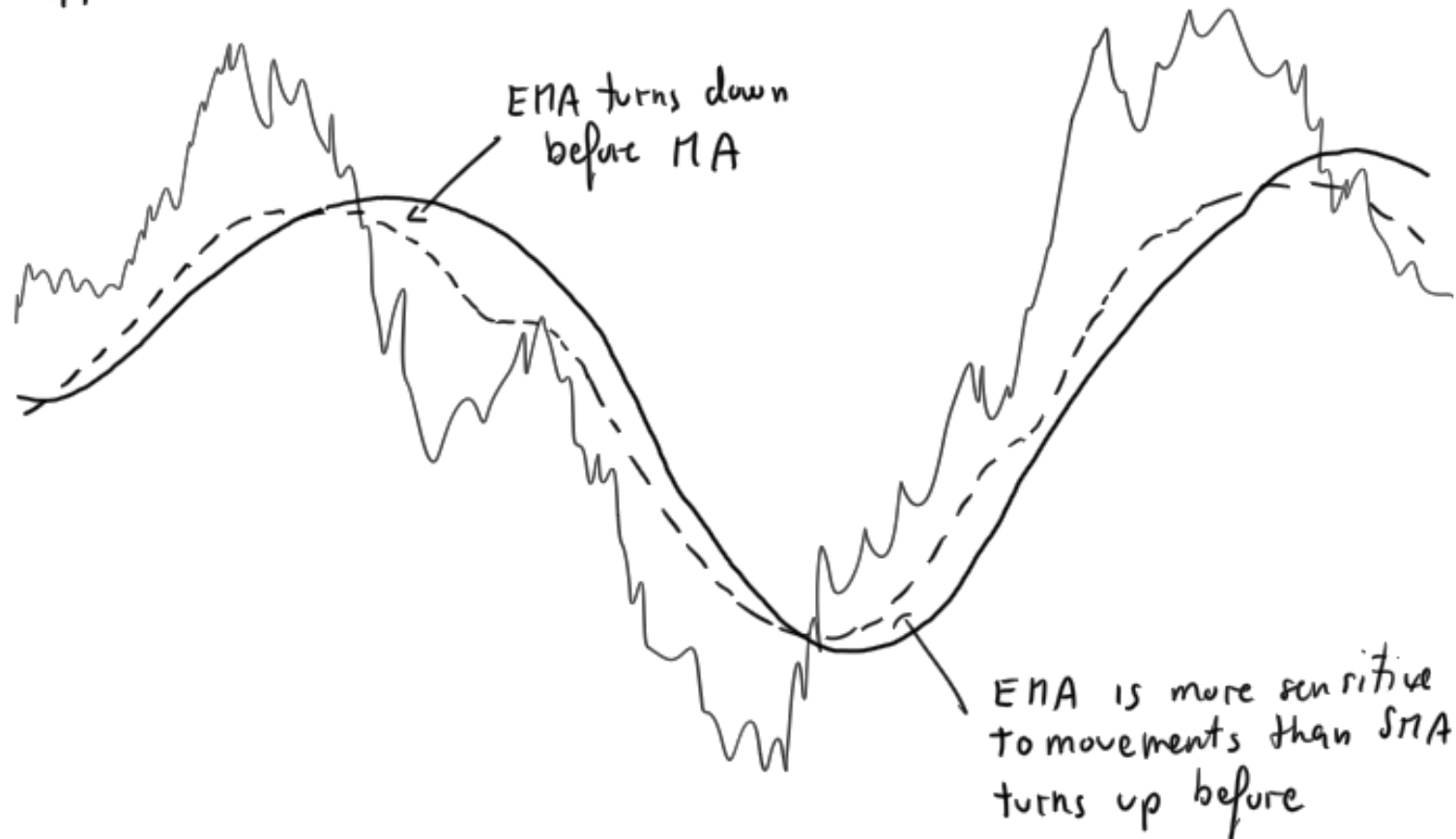
Image by Kerstin Riemer from Pixabay

Modeling traditional techniques

Capturing trend via smoothing

MA —

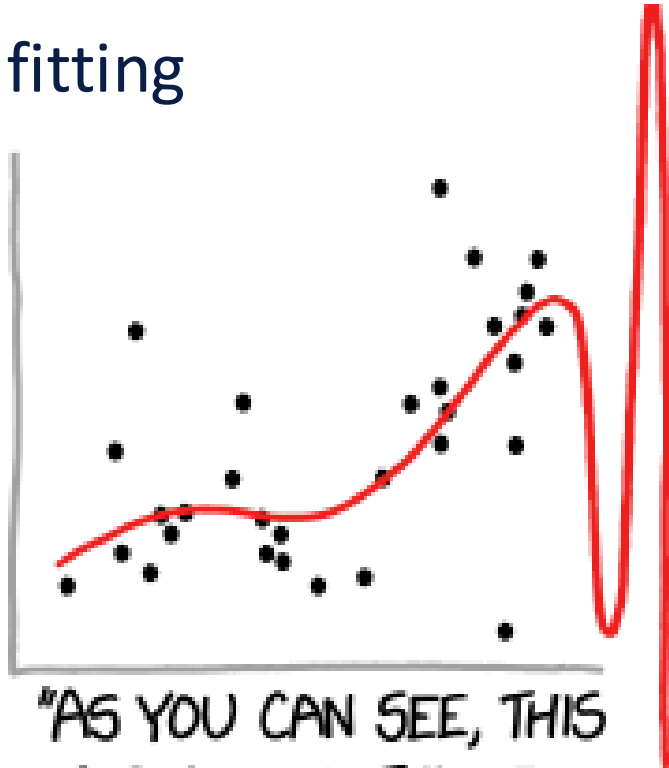
ENA ---



Modeling traditional techniques

Capturing the Trend

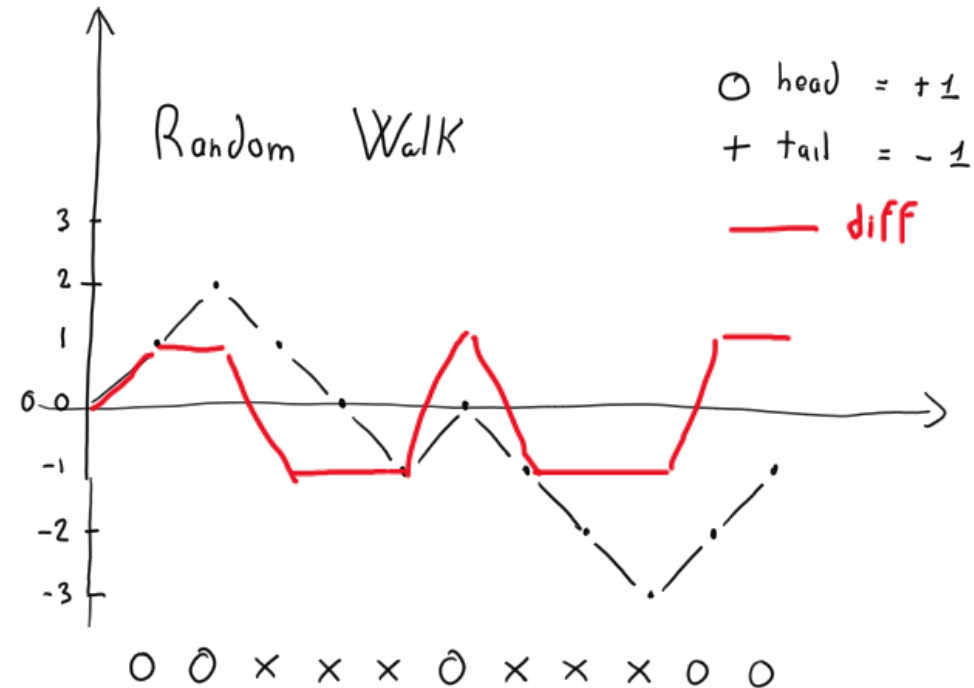
Not fitting



"AS YOU CAN SEE, THIS
MODEL SMOOTHLY FITS
THE— WAIT NO NO DON'T
EXTEND IT AAAAAA!!"

[src:; <https://xkcd.com/2048/>]

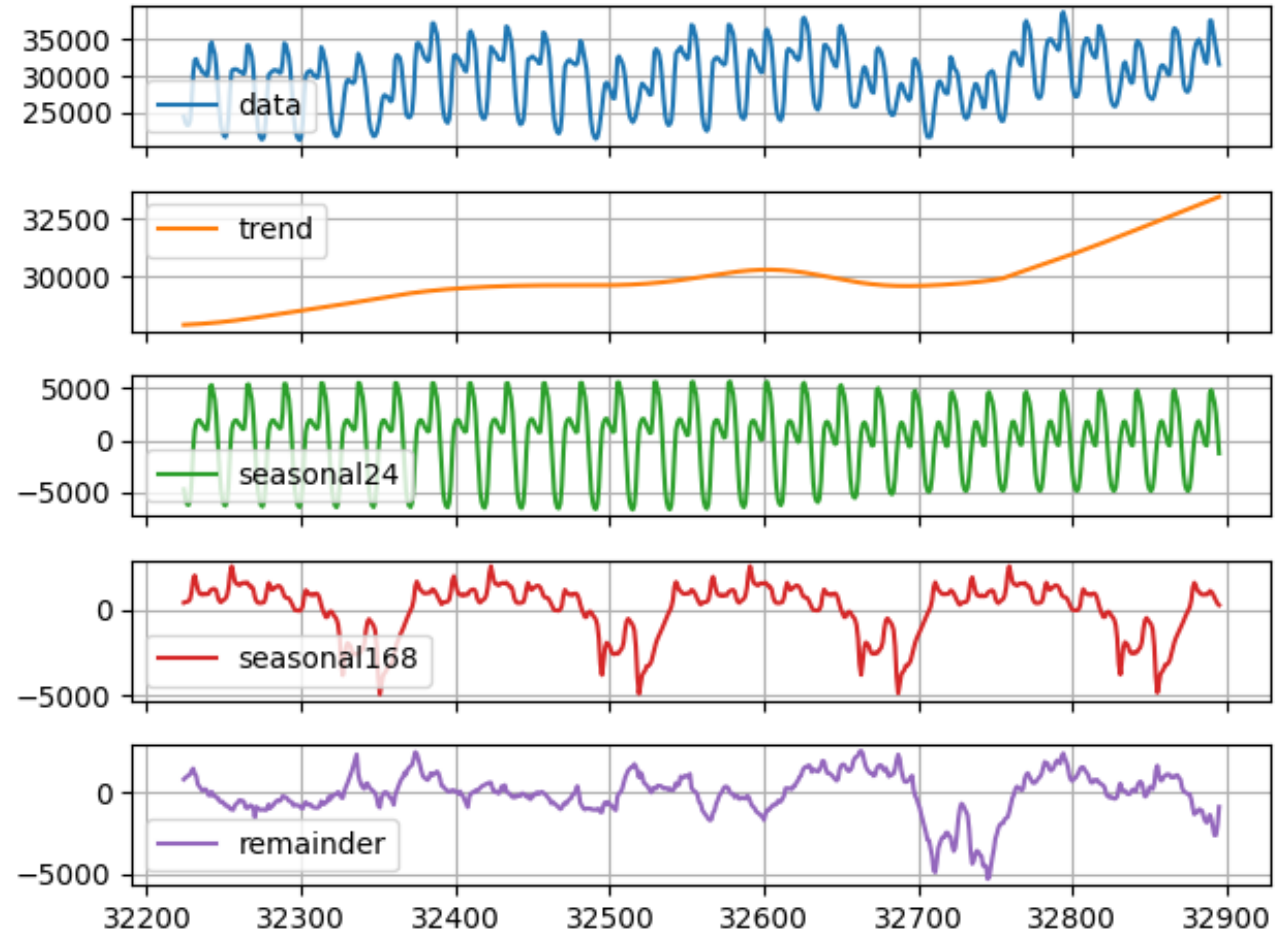
... but smoothing or differencing



Modeling traditional techniques

Capturing Seasonality

- Multiple seasons are typically **present in data sampled at low frequency**.
- E.g., **hourly electricity data** exhibits daily and weekly seasonality, i.e., there are clear **patterns** of electricity consumption for specific
 - **hours** of the day like 6:00pm vs 3:00am
 - **days** like Sunday vs Friday.



Forecasting traditional techniques

Autoregressive moving-average (ARMA)

- Autoregressive

$$X_t = \sum_{i=1}^p \varphi_i X_{t-i} + \varepsilon_t$$

- Moving-average

$$X_t = \mu + \varepsilon_t + \sum_{i=1}^q \theta_i \varepsilon_{t-i}$$

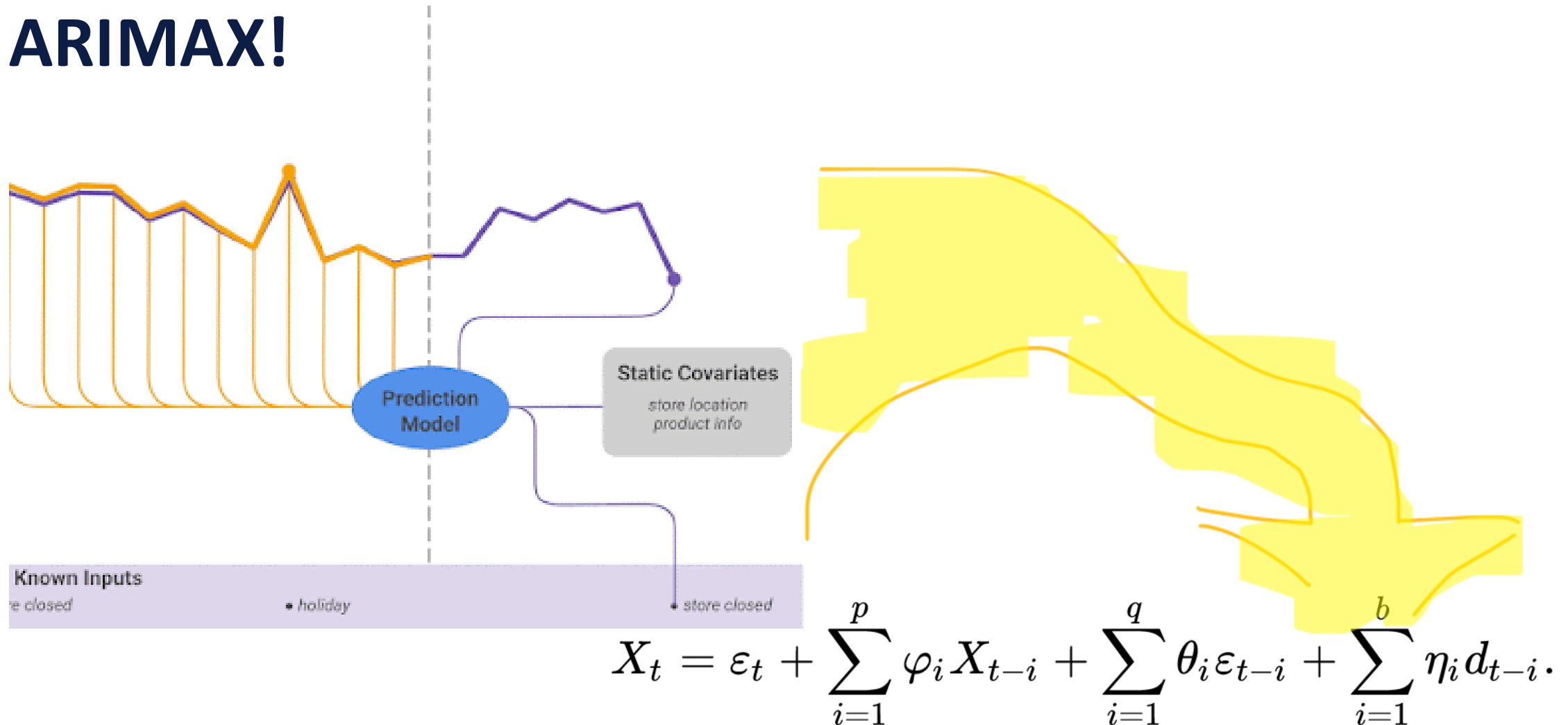
- ARMA

$$X_t = \varepsilon_t + \sum_{i=1}^p \varphi_i X_{t-i} + \sum_{i=1}^q \theta_i \varepsilon_{t-i}.$$

Forecasting traditional techniques

What about using external data (covariates)?

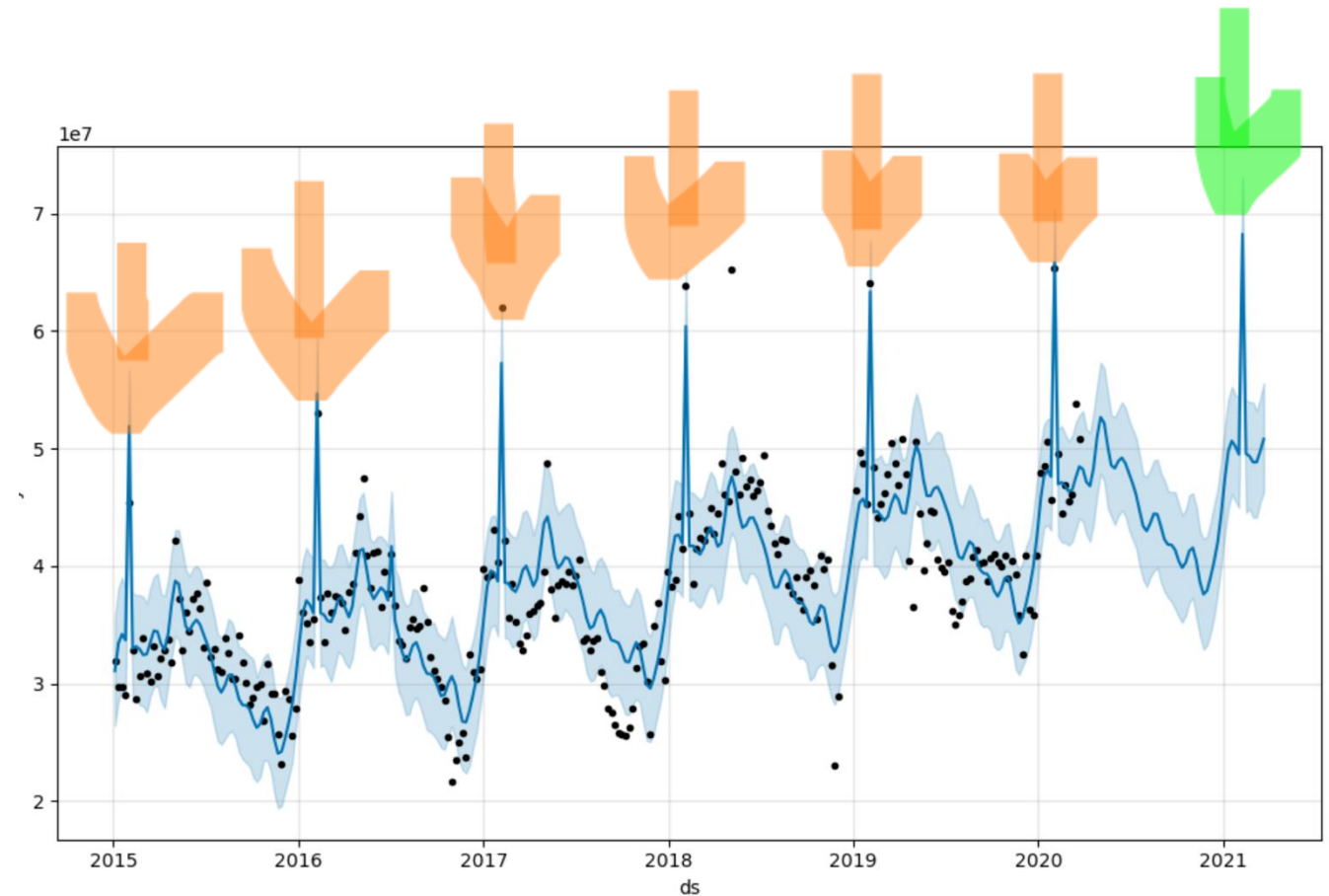
ARIMAX!



Forecasting traditional techniques

Meta's Prophet

- Completely automatic forecasting techniques can be brittle and often too rigid to incorporate useful assumptions or heuristics.
- Prophet is essentially “framing the forecasting problem as a curve-fitting exercise” rather than looking explicitly at the time based dependence of each observation.

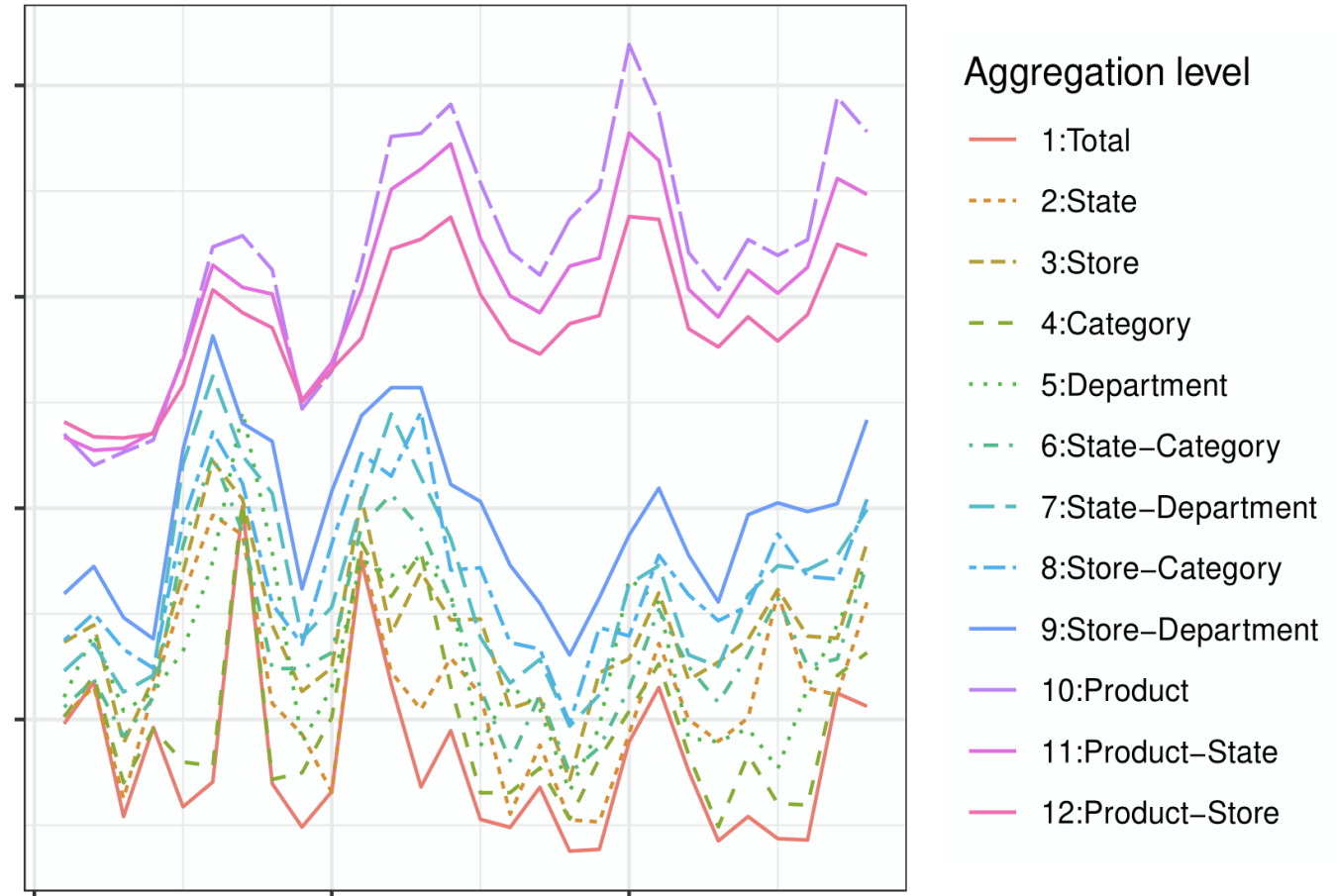


- E.g., holidays effects

Challenging traditional methods

Unit sales forecasting for 1000s of products

- Forecasting the unit sales of 3,049 Walmart products grouped by type (category and department) and selling location (stores and states).
- E.g., Makridakis's challenge <https://mofc.unic.ac.cy/m5-competition/>

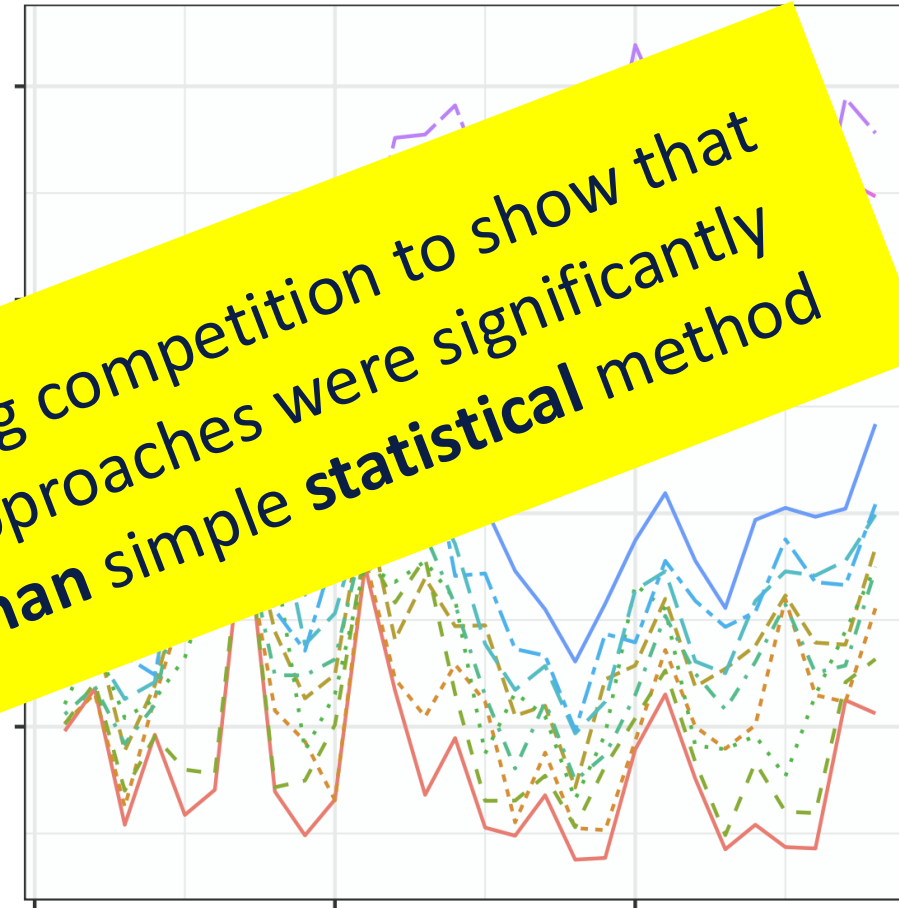


Challenging traditional methods

Unit sales forecasting for 1000s of products

- Forecasting the unit sales of 3,049 Walmart products grouped by type (category and department) and selling location (stores and states)
- E.g., Makridakis' <https://mofc.uconn.edu/>

The first forecasting competition to show that two ML-based approaches were significantly more accurate than simple **statistical** method



Aggregation level

- 1:Total
- - - 2:State
- . - 3:Store
- - - 4:Category
- ... 5:Department
- . - 6:State-Category
- - - 7:State-Department
- . - 8:Store-Category
- 9:Store-Department
- 10:Product
- 11:Product-State
- 12:Product-Store



Advanced techniques Specialized Neural Network

N-BEATS: NEURAL BASIS EXPANSION ANALYSIS FOR INTERPRETABLE TIME SERIES FORECASTING

Boris N. Oreshkin
Element AI
boris.oreshkin@gmail.com

Dmitri Carpov
Element AI
dmitri.carpov@elementai.com

Nicolas Chapados
Element AI
chapados@elementai.com

Yoshua Bengio
Mila
yoshua.bengio@mila.quebec

ABSTRACT

We focus on solving the univariate times series point forecasting problem using deep learning. We propose a deep neural architecture based on backward and forward residual links and a very deep stack of fully-connected layers. The architecture has a number of desirable properties, being interpretable, applicable without modification to a wide array of target domains, and fast to train. We test the proposed architecture on several well-known datasets, including M3, M4 and TOURISM competition datasets containing time series from diverse domains. We demonstrate state-of-the-art performance for two configurations of N-BEATS for

DeepAR: Probabilistic Forecasting with Autoregressive Recurrent Networks

David Salinas, Valentin Flunkert, Jan Gasthaus
Amazon Research
Germany
<dsalina,flunkert,gasthaus@amazon.com>

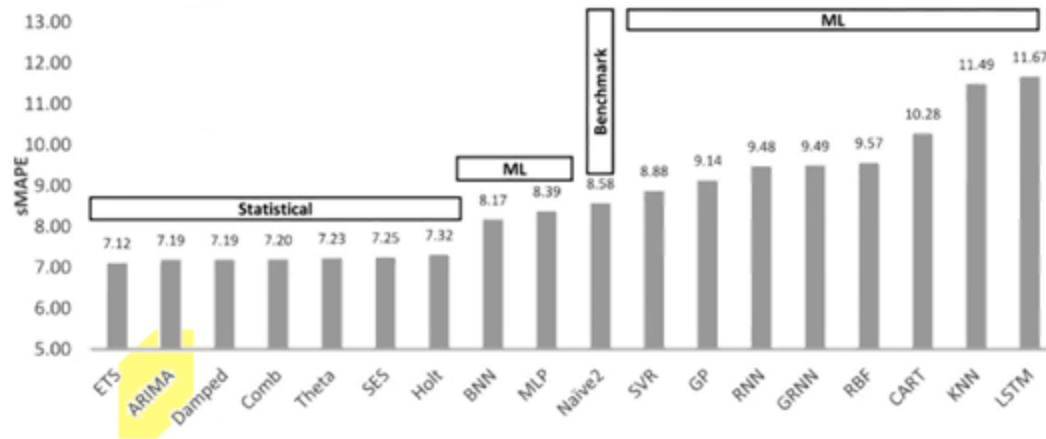
Abstract

Probabilistic forecasting, i.e. estimating the probability distribution of a time series' future given its past, is a key enabler for optimizing business processes. In retail businesses, for example, forecasting demand is crucial for having the right inventory available at the right time at the right place. In this paper we propose DeepAR, a methodology for producing accurate probabilistic forecasts, based on training an auto-regressive recurrent network model on a large number of related time series. We demonstrate how by applying deep learning techniques to forecasting, one can overcome many of the challenges faced by widely-used classical approaches to the problem. We show through extensive empirical evaluation on several real-world forecasting data sets accuracy improvements of around 15%.

Be aware of trade-offs!

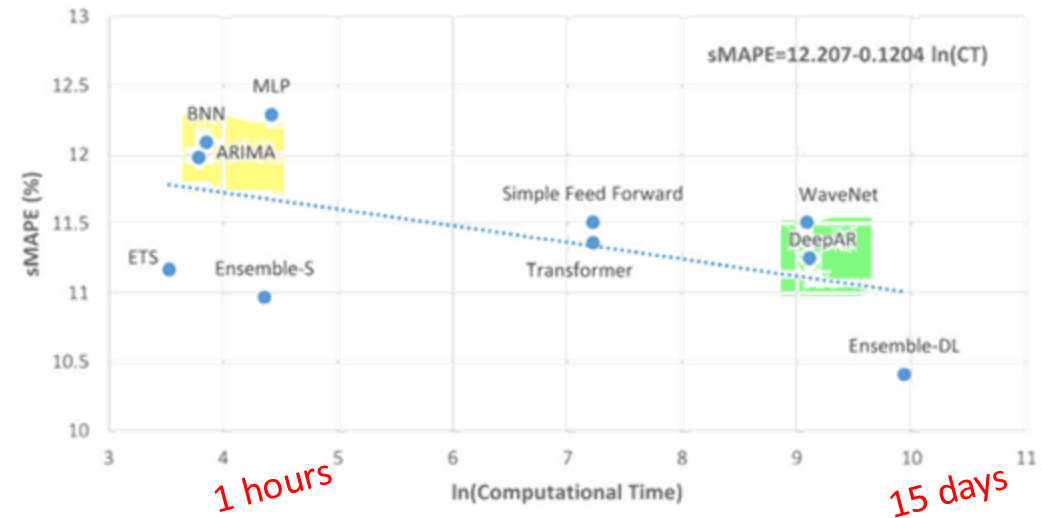
2018

Statistical methods largely outperform ML-based methods



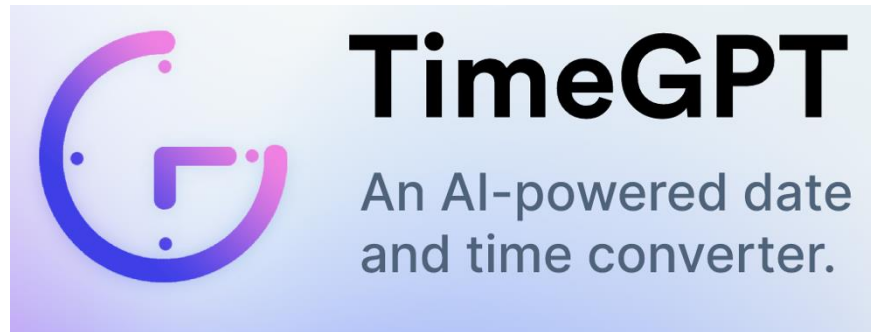
2022

ML-based methods (i.e., deep learning ones) outperform statistical methods *but at a vast computational cost*



The advent of the foundational models

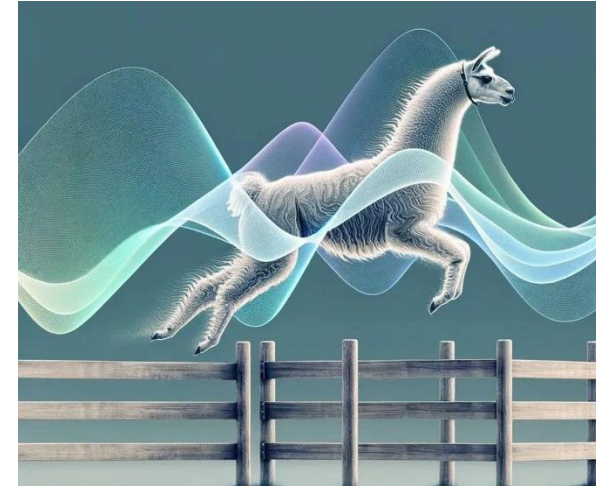
Nixtla's



<https://www.nixtla.io/timegpt>

Amazon's
CHRONOS

<https://github.com/amazon-science/chronos-forecasting>



Meta's Lag-Llama

<https://github.com/time-series-foundation-models/lag-llama>

Google's TimeFM

<https://github.com/google-research/times>



Learning Characteristics

- Time series analytics is not inherently adaptive nor focused on real-time learning
- Models are (almost always) retrained from scratch as new data points are received.

Learning one data at a time
from a continuous flow of data
without assuming identically distributed data
with
Streaming Machine Learning



Type of data

data arrives in a **continuous stream**, potentially in real-time, assuming **data** points are **independent but** data stream can evolve over time (i.e., the data are **not identically distributed**)

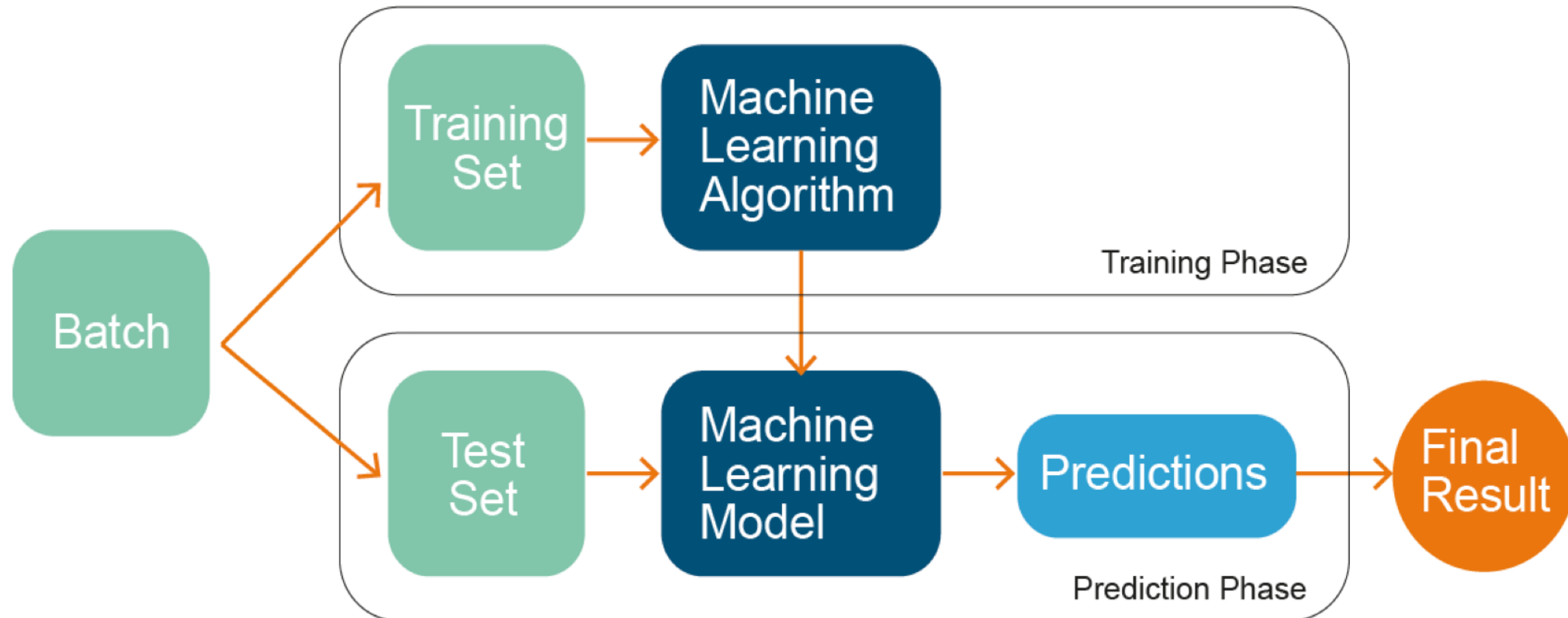
Purpose

- extract **insights** from data **as** it arrives and discard the data
- often with
 - a low-latency requirement
 - a minimal computational footprint
- classification is the main application
- forecasting and clustering are also in scope

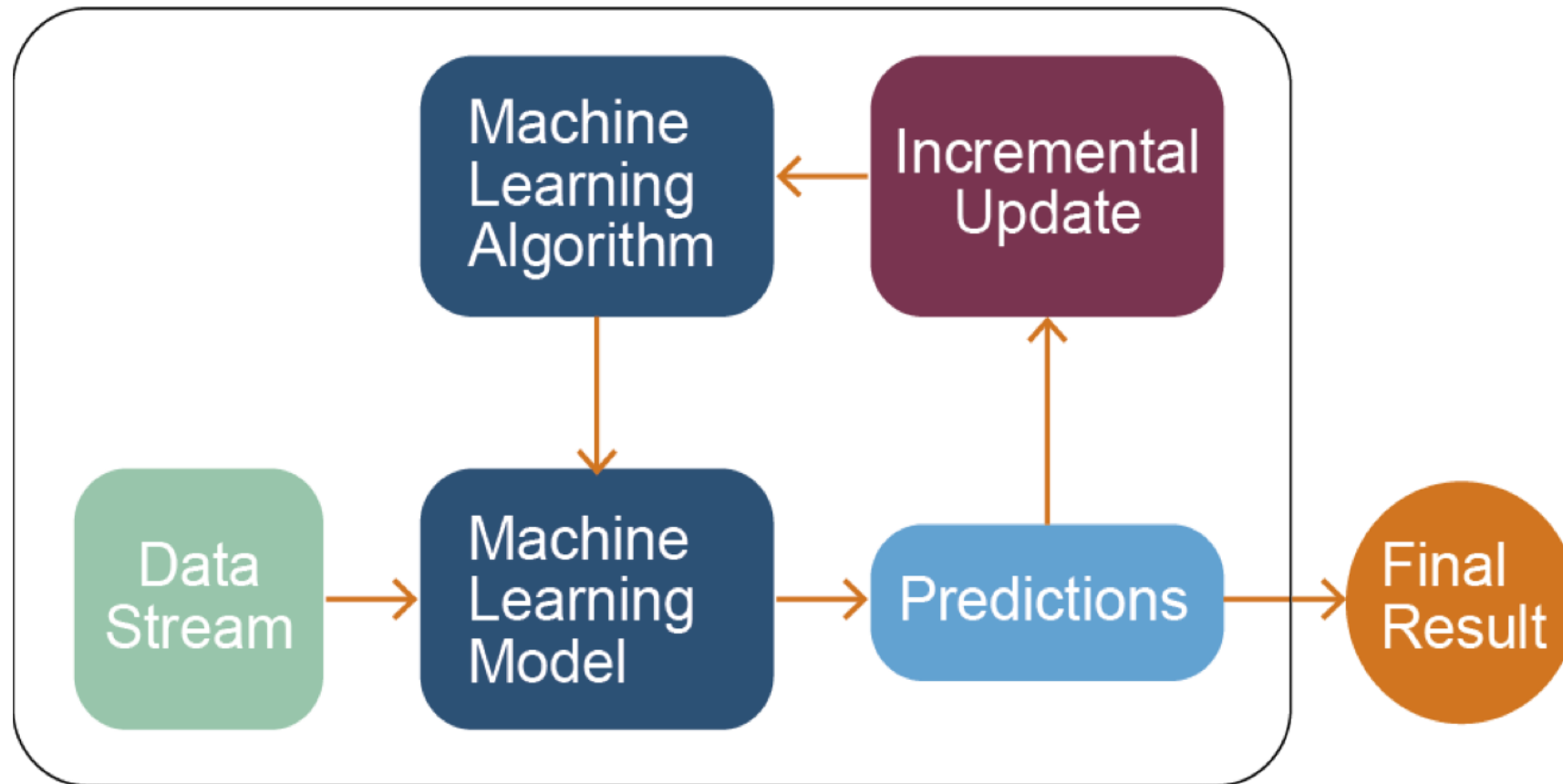


[src: <http://www.motusml.com>]

ML models



SML models

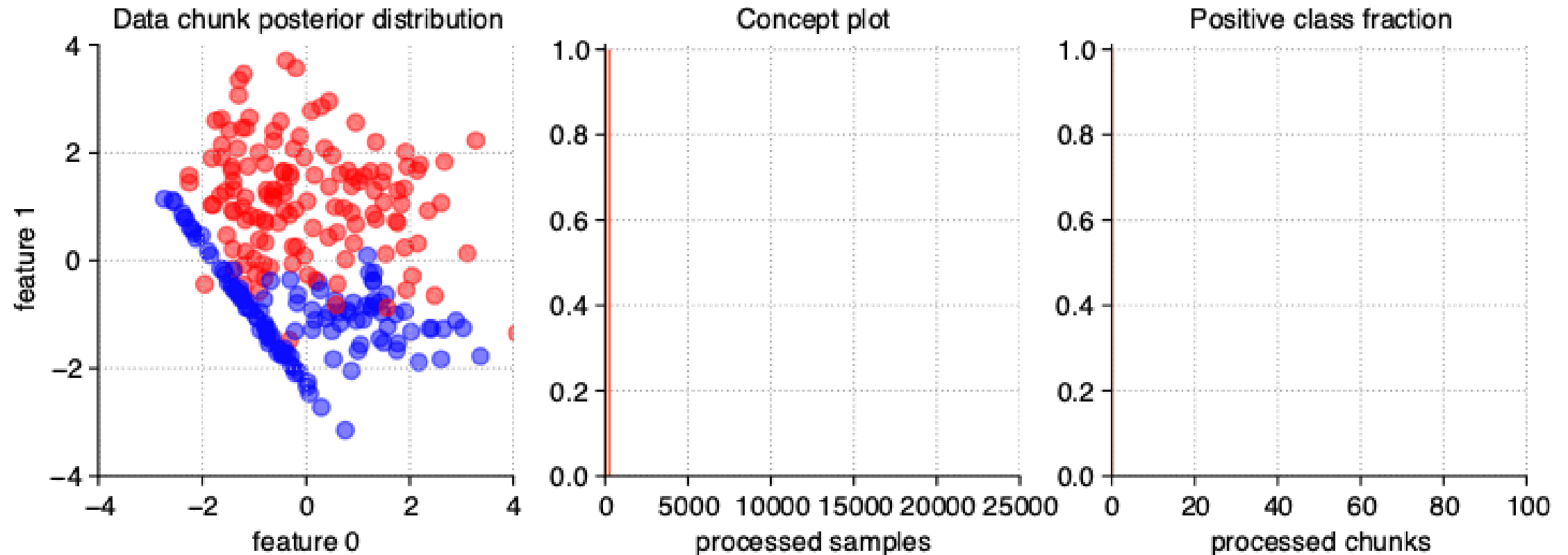


A. Bifet, G. de Francisci Morales, J. Read, G. Holmes, and B. Pfahringer **Efficient online evaluation of big data stream classifiers**. ACM SIGKDD 2015

The basics

Data arrives in a continuous data stream

balanced data or



[src: https://stream-learn.readthedocs.io/en/latest/_images/stationary.gif]

Incremental Techniques (a.k.a. Online ML)

Hoeffding Trees

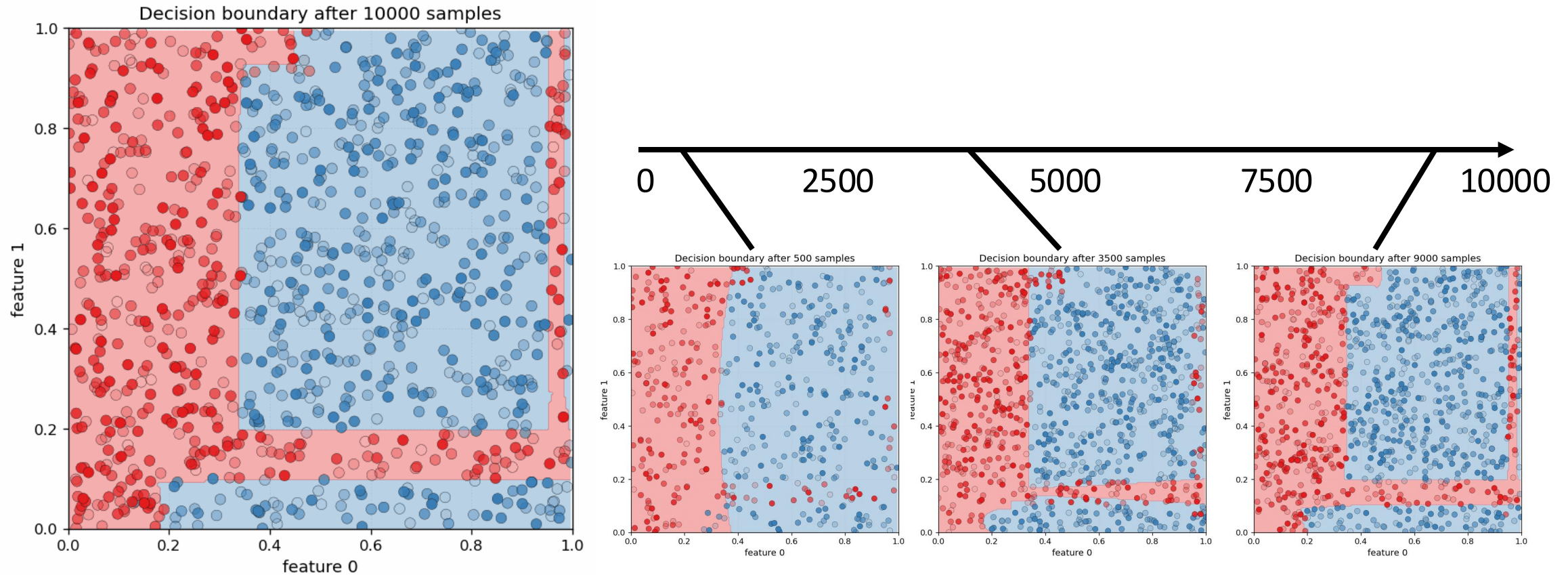
- **Decision Tree built incrementally**
 - A data point at a time
 - Memory only stores the model
- There are **theoretical guarantees** that the final tree is identical (with high probability) to a tree built using a batch decision tree algorithm
- Split if $H(x_1) - H(x_2) > \varepsilon = \sqrt{\frac{R^2 * \log(1/\delta)}{2N}}$



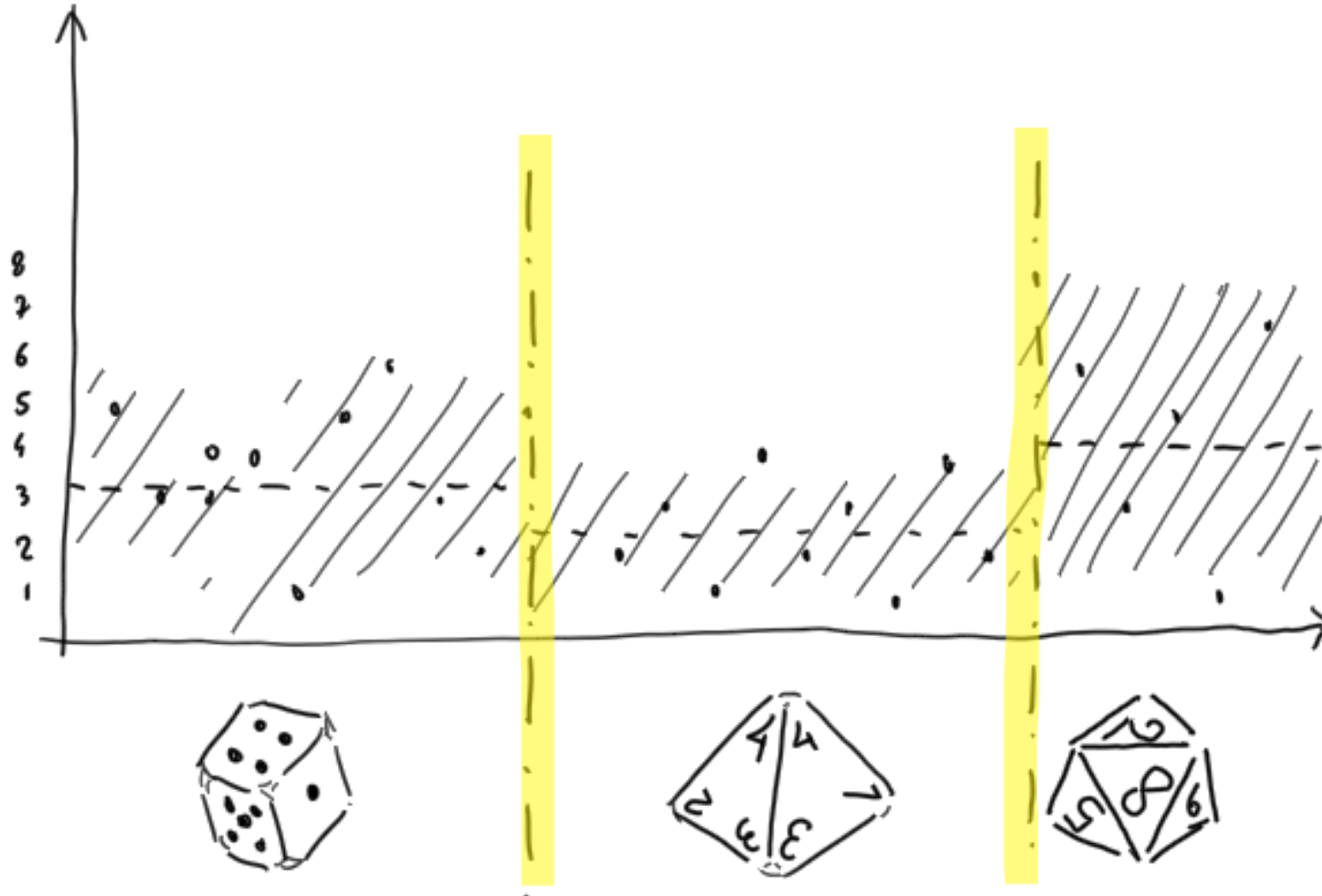
[src: <https://www.bilibili.tv/en/video/4787736782838275>]

Incremental Techniques (a.k.a. Online ML)

How does an Hoeffding Trees learn?



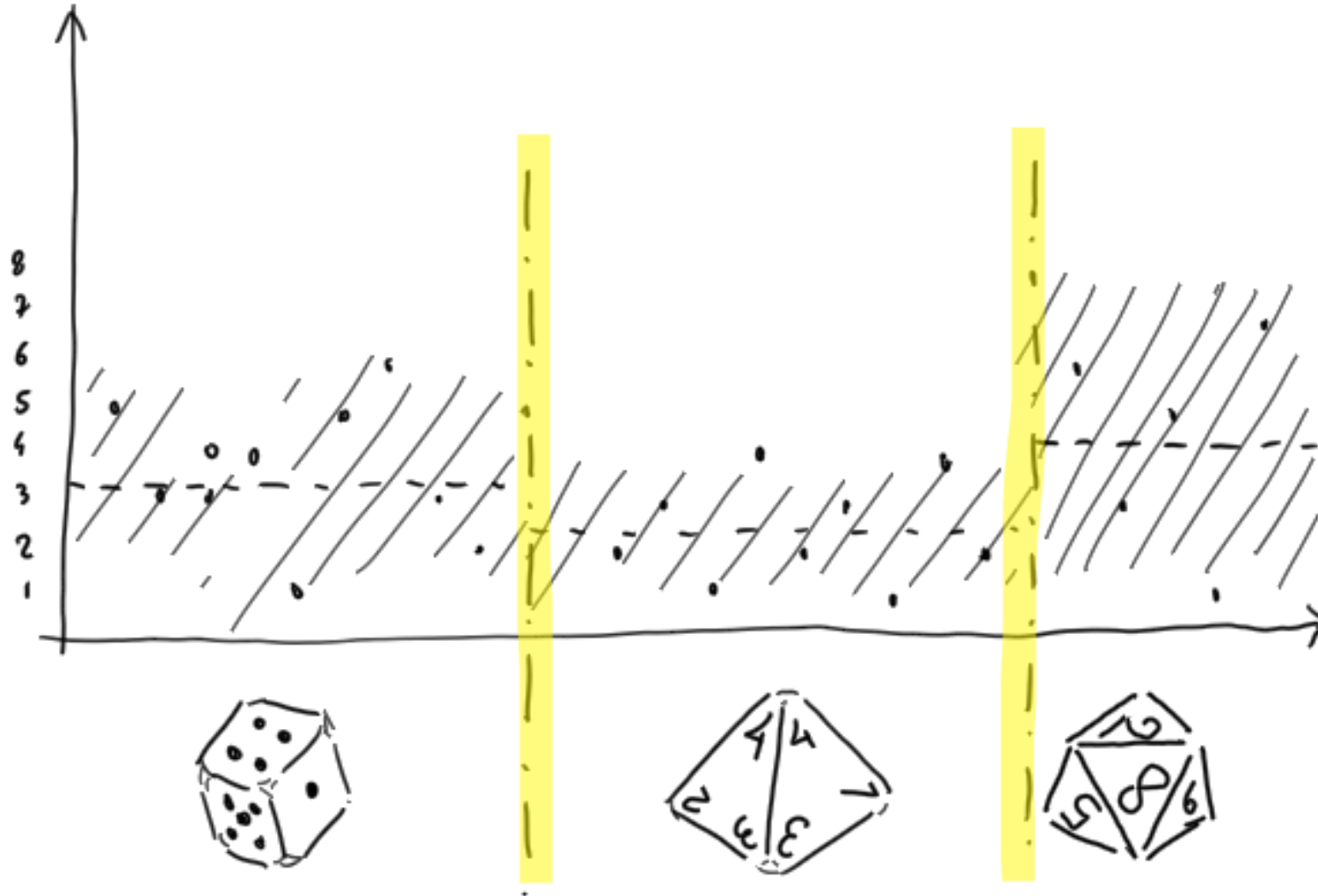
Recall: Non identically distributed data



DRIFTS
(a.k.a., non-stationary in TSA
temporal distribution shift in ML)

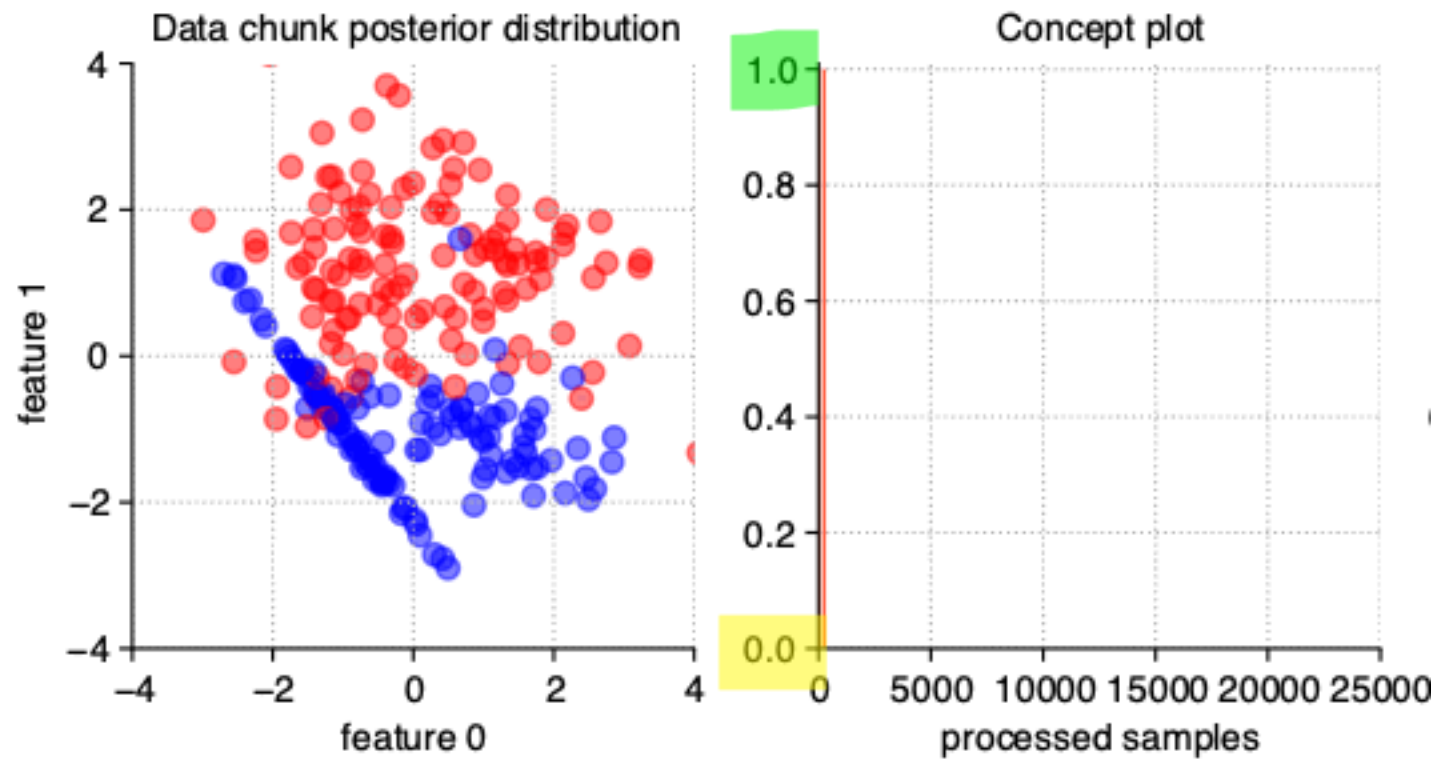
Precisely, a Data Drift (i.e., a change in X^*)

* X is the input distribution



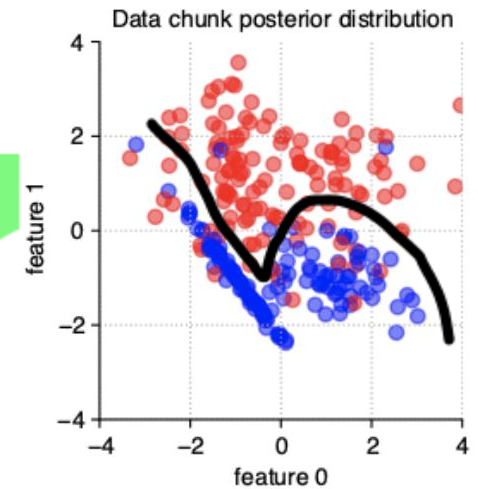
DATA DRIFTS
(a.k.a., Virtual Concept Drift)

Concept Drift is a change in $X \rightarrow y$!!!

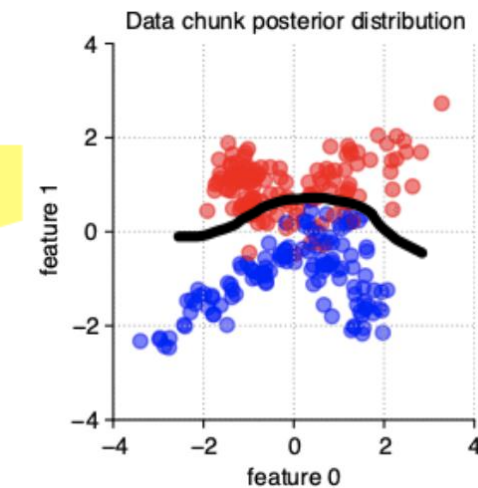


[src: https://stream-learn.readthedocs.io/en/latest/_images/incremental.gif]

Concept 1



Concept 0

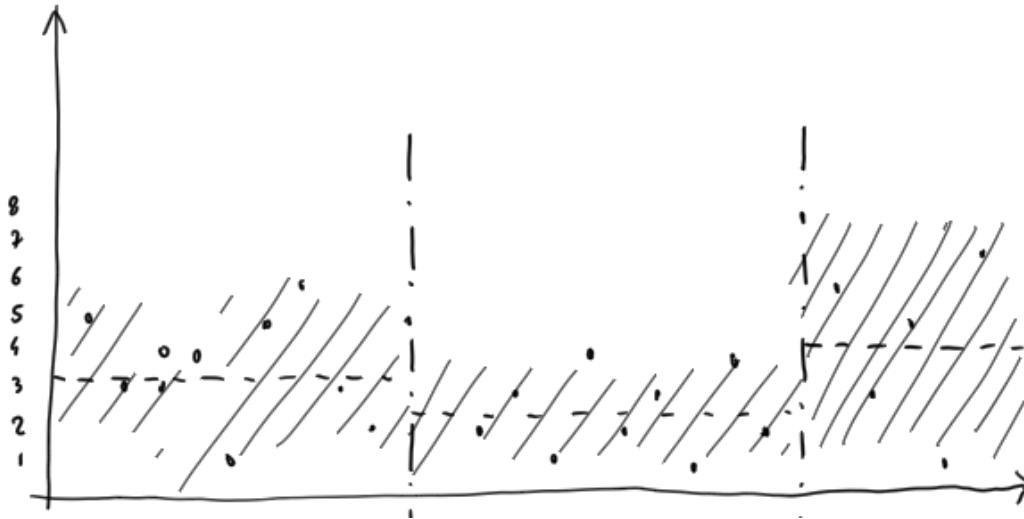


Techniques

Drift detectors

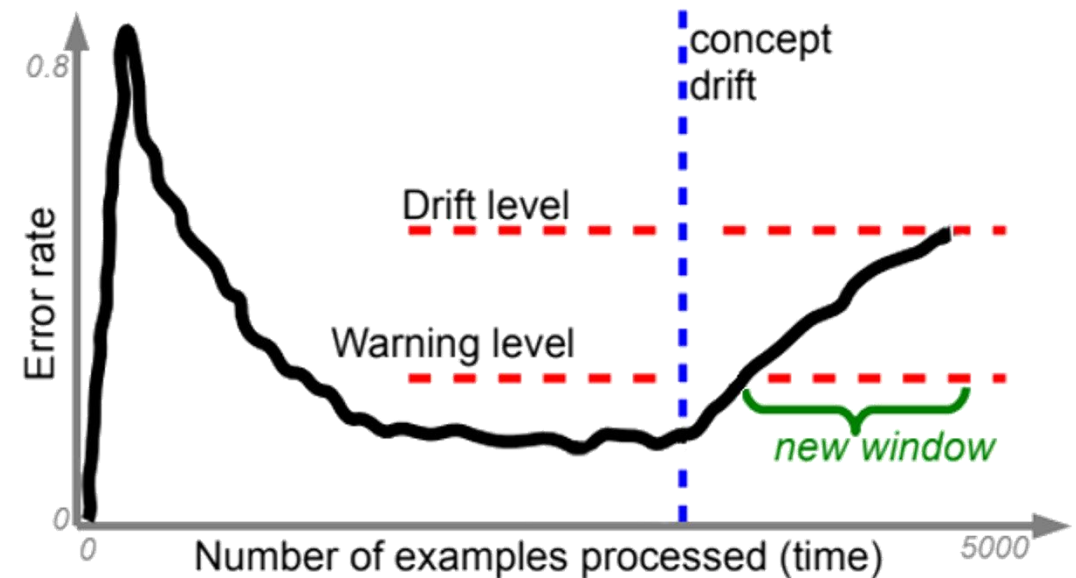
Monitoring the input distribution

Is there a statistically significant **difference between** the distribution of the **recent X** and the one of the **old X**?



Monitoring the prediction error

Is there a statistically significant **growth** between the recent and old **errors**?

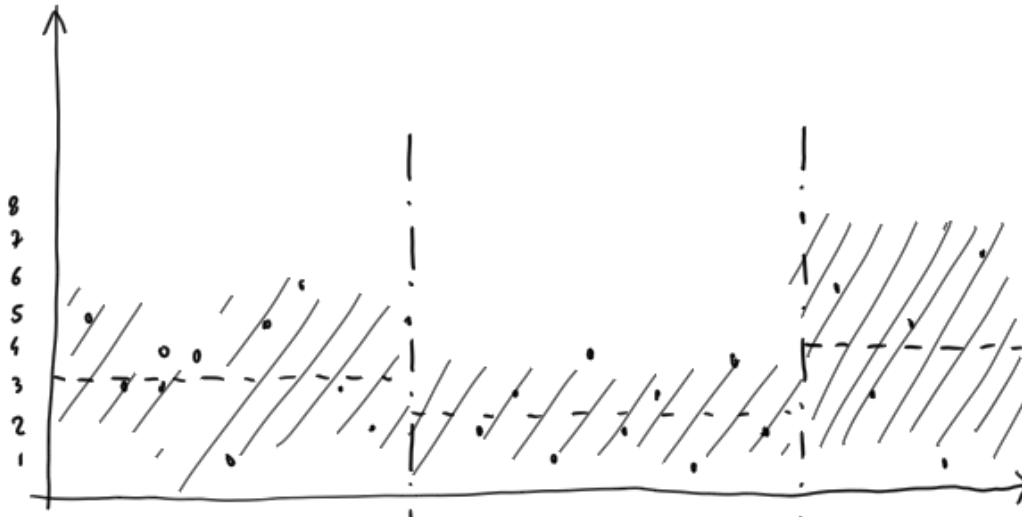


Techniques

Drift detectors

Monitoring the input distribution

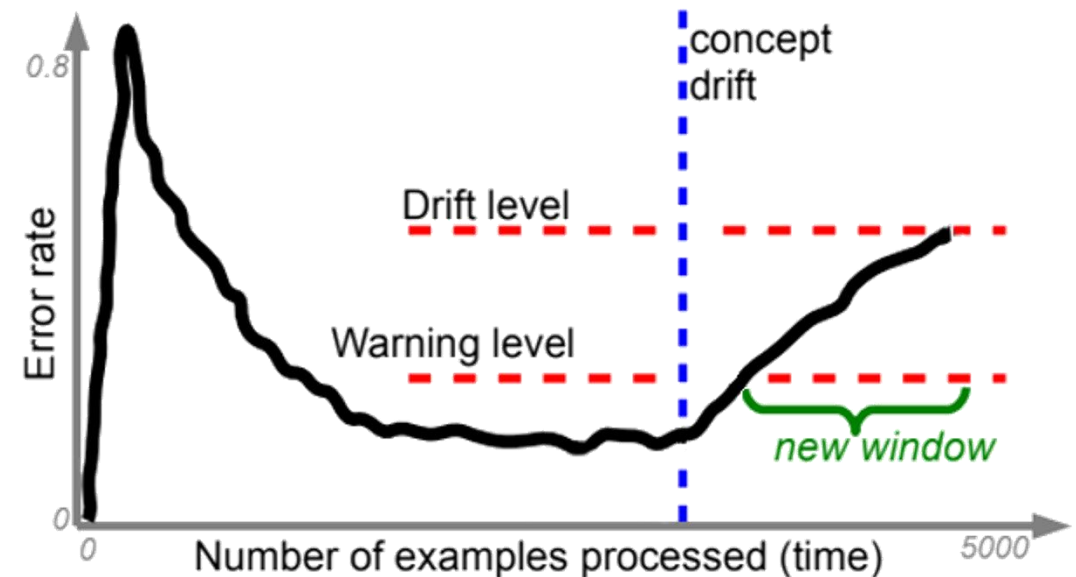
Is there a statistically significant **difference between** the distribution of the **recent X** and the one of the **old X**?



ADWIN!

Monitoring the prediction error

Is there a statistically significant **growth** between the recent and old **errors**?



Streaming Machine Learning Techniques

Hoeffding Adaptive Tree (HAT)

PROBLEM:

- When a **concept drift** occurs, part of the decision tree may **lose accuracy**

SOLUTION:

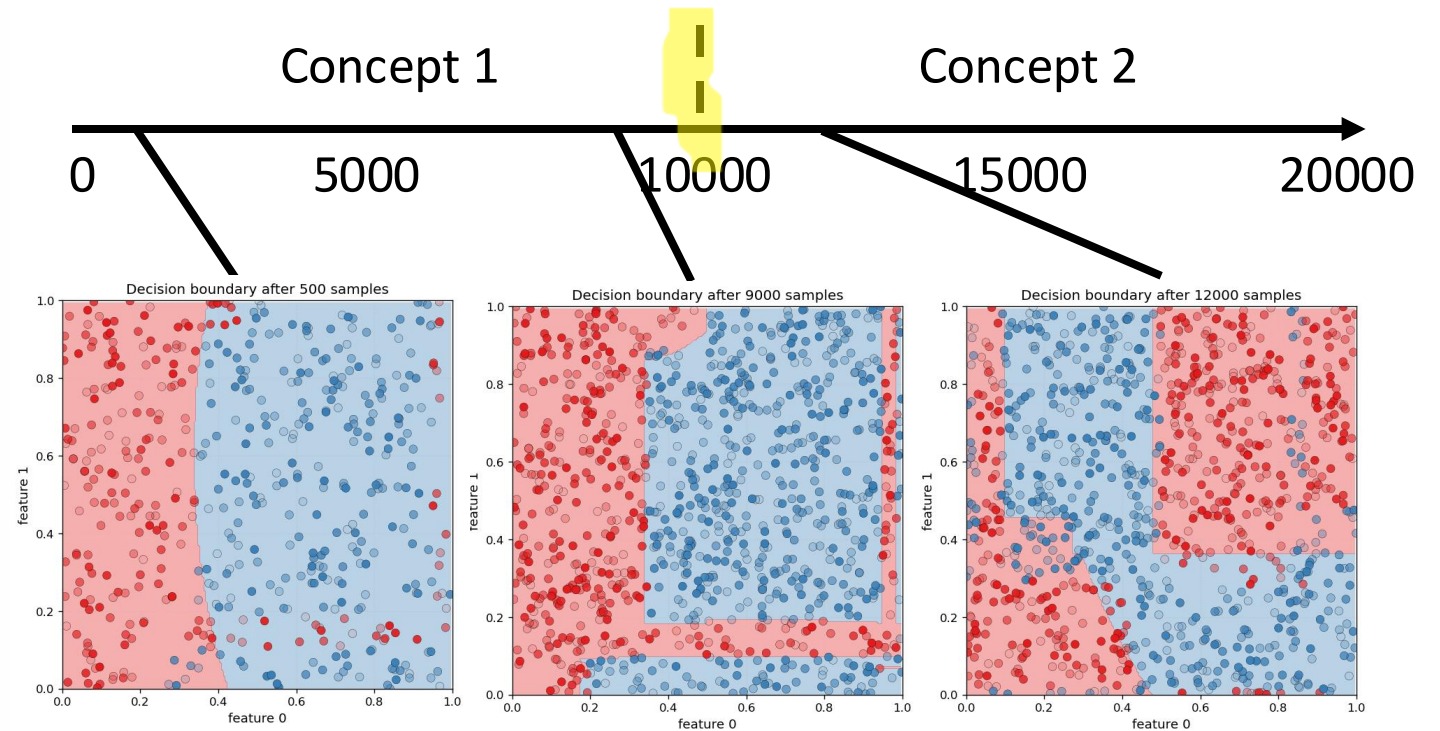
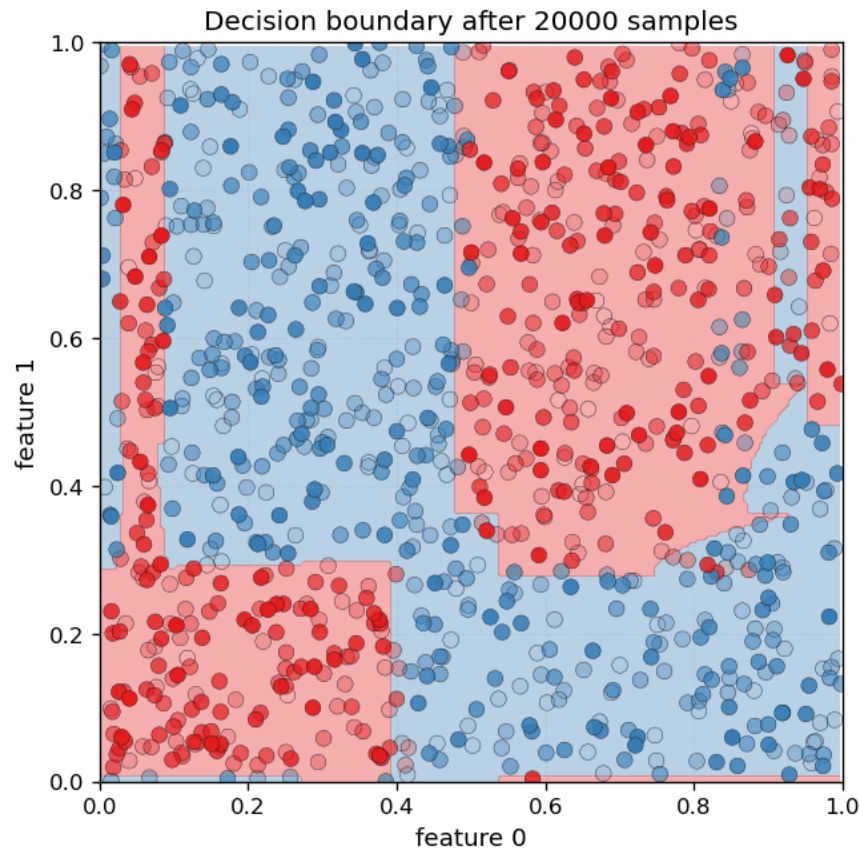
- When **ADWIN** detects a concept drift, start **growing alternative branches**
- **When** they become **more accurate** than existing ones, **switch them** with the old ones



[src: <https://giphy.com/gifs/dance-dancing-groot-jwTqLNfrx4OPe>]

Streaming Machine Learning Techniques

How does an Hoeffding Adaptive Trees learn?



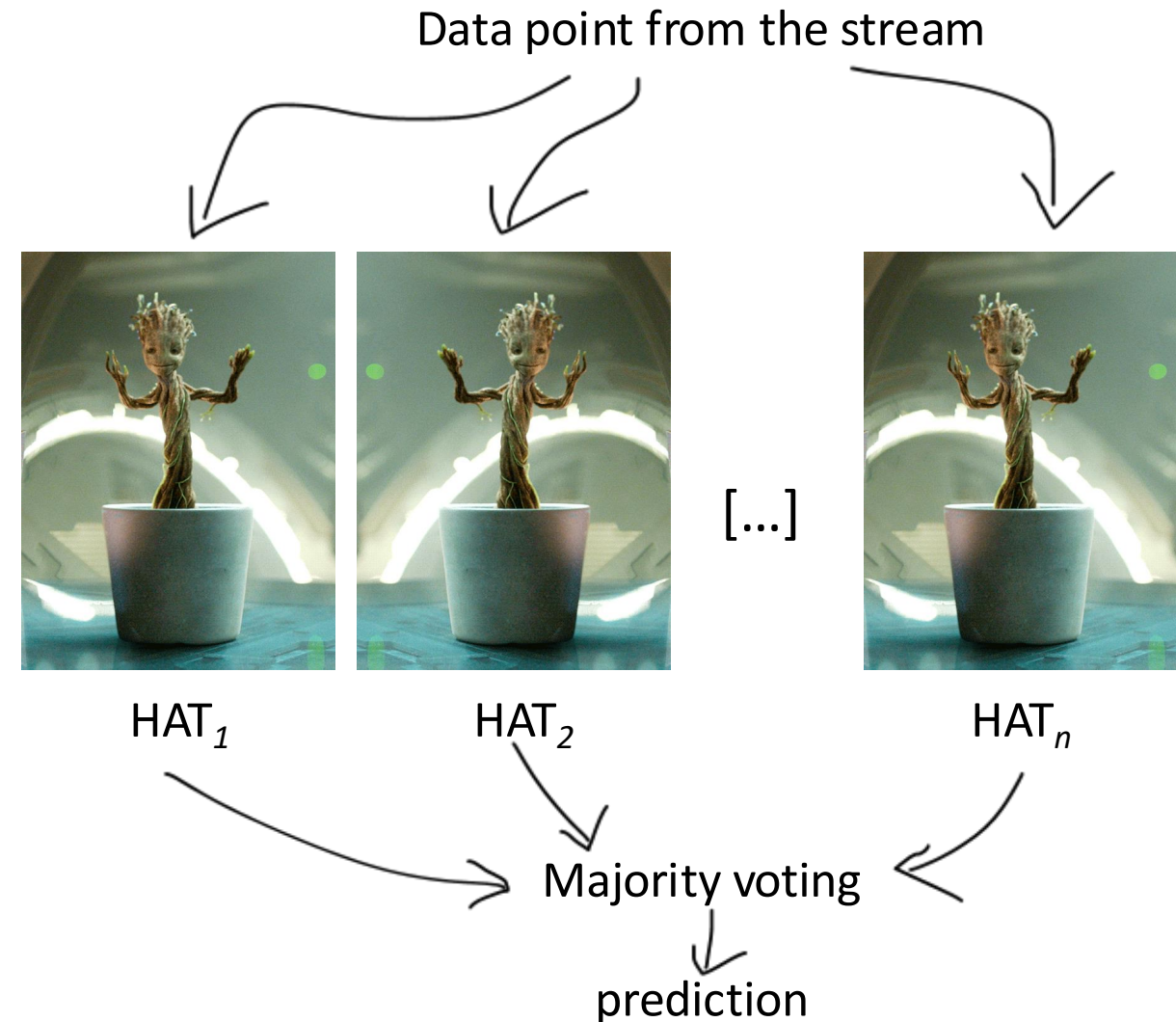
Techniques Ensembles!

Via **Bagging**

- **Combine n HAT** to produce an output with a lower variance
- E.g., **Adaptive Random Forest**
(see animation to the right)

Via **Boosting**

- Fits HAT models **sequentially** on the **residuals** of the previous HAT model
- E.g., **Adaptive Boosting**



Learning Characteristics

Streaming **M**achine **L**earning is

- **adaptive** and designed to handle
 - changing data distributions, a.k.a., data drifts or virtual concept drift
 - Concept drifts, a.k.a., changes in $X \rightarrow y$
- well-suited for applications requiring **immediate response** to incoming data **changes**



[src: <https://gifer.com/en/2rci>]

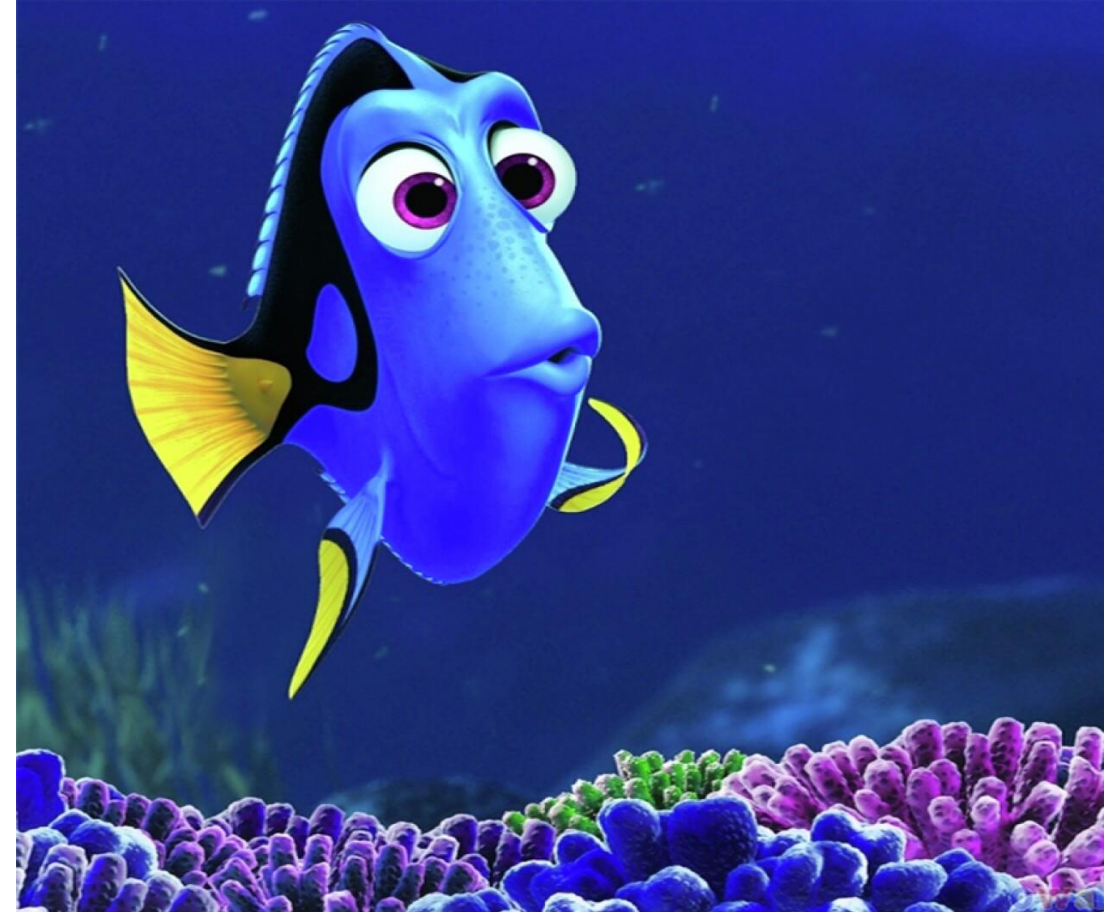
Reflection

QUESTION

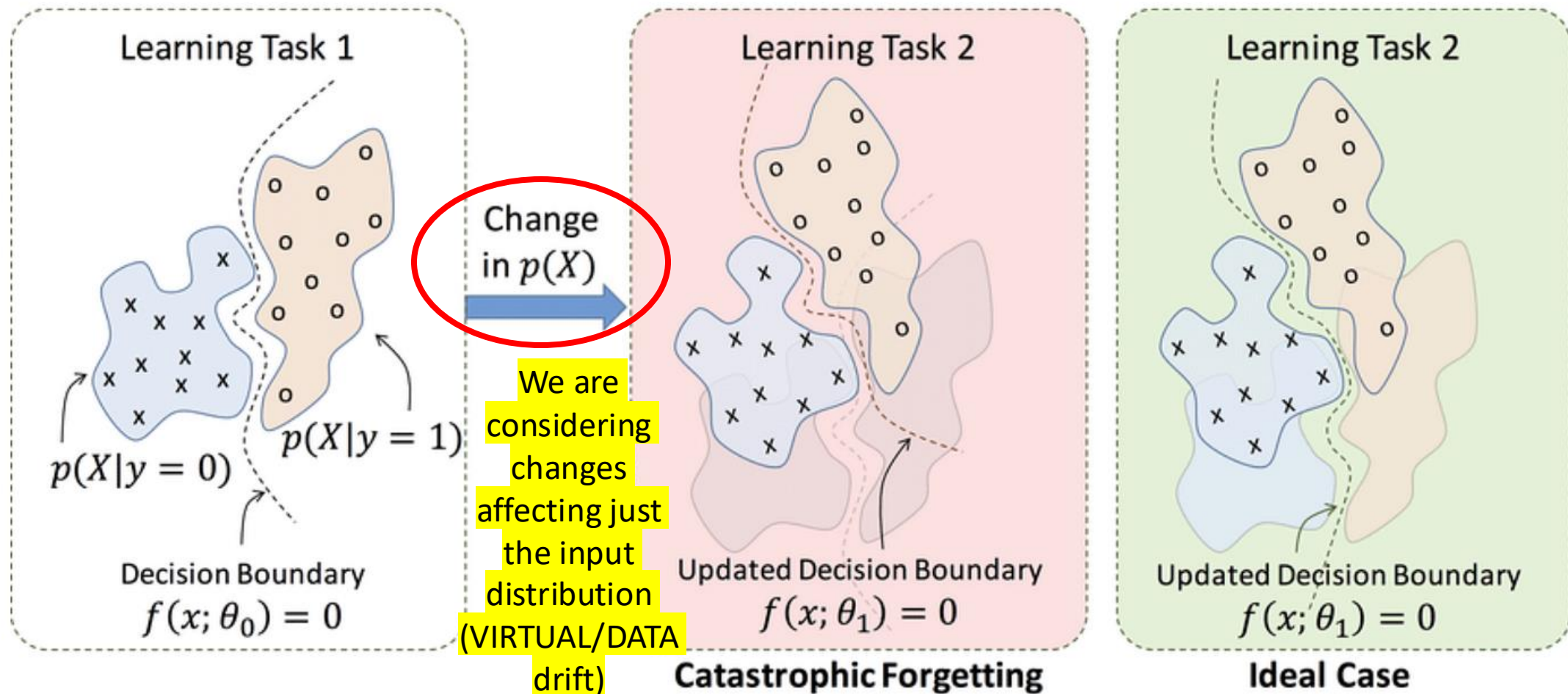
- At what cost does SML immediately react to changes?

ANSWER

- It forgets!



Catastrophic forgetting



[src: <https://arxiv.org/pdf/1903.06070.pdf>]

The plasticity-stability dilemma in AI*

Plasticity is the ability to
learn **new** knowledge



Photo by Antoni Shkraba

Stability is the ability to
remember past knowledge

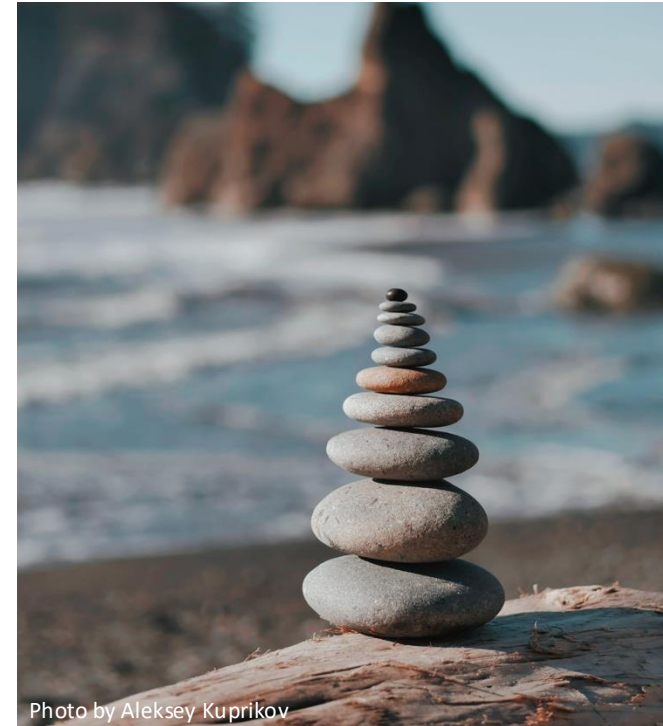


Photo by Aleksey Kuprikov

* McCloskey & Cohen, 1989 - <https://www.andywills.info/hbab/mccloskeycohen.pdf>

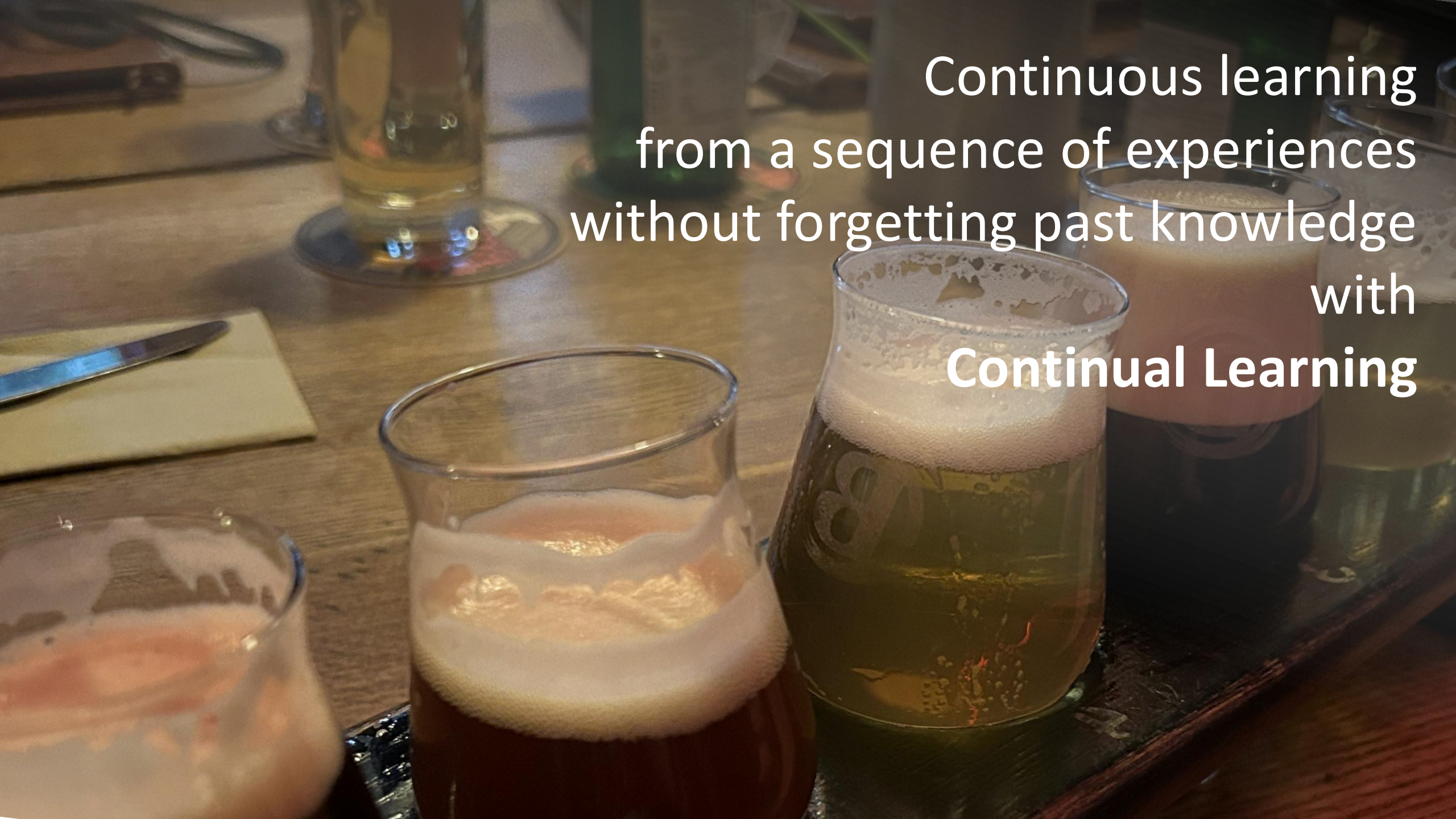
The plasticity-stability dilemma in AI*

Too much stability leads in difficulties in learning



Too much plasticity leads to the catastrophic **forgetting** problem

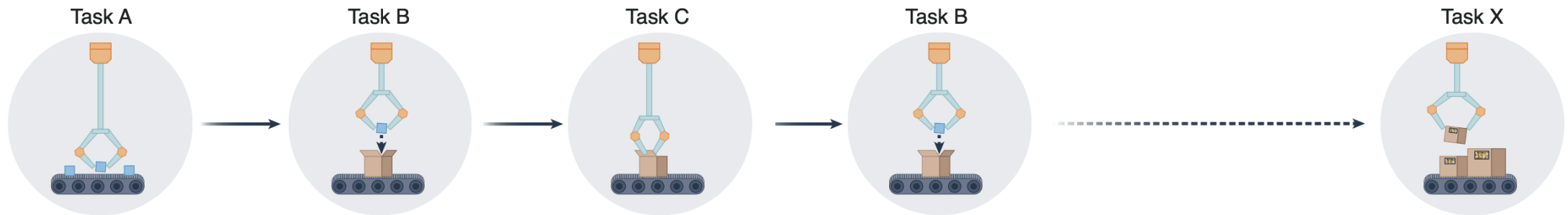


A photograph of a wooden table in a bar or pub setting. In the foreground, there are several glasses of beer. One glass on the left has a thick, white head of foam. Next to it is a glass with a dark beer and a thick, white head of foam. To the right of that is a glass with a golden beer and a thick, white head of foam. Further back, there is another glass with a golden beer and a thick, white head of foam. On the left side of the table, there is a yellow napkin with a knife resting on it. The background is slightly blurred, showing more of the bar area.

Continuous learning
from a sequence of experiences
without forgetting past knowledge
with
Continual Learning

Continual learning definition & an example

- Learning from a sequence of training episodes intermixed with situations that require applying (recently or previously) learned skills
- E.g., a robotic arm

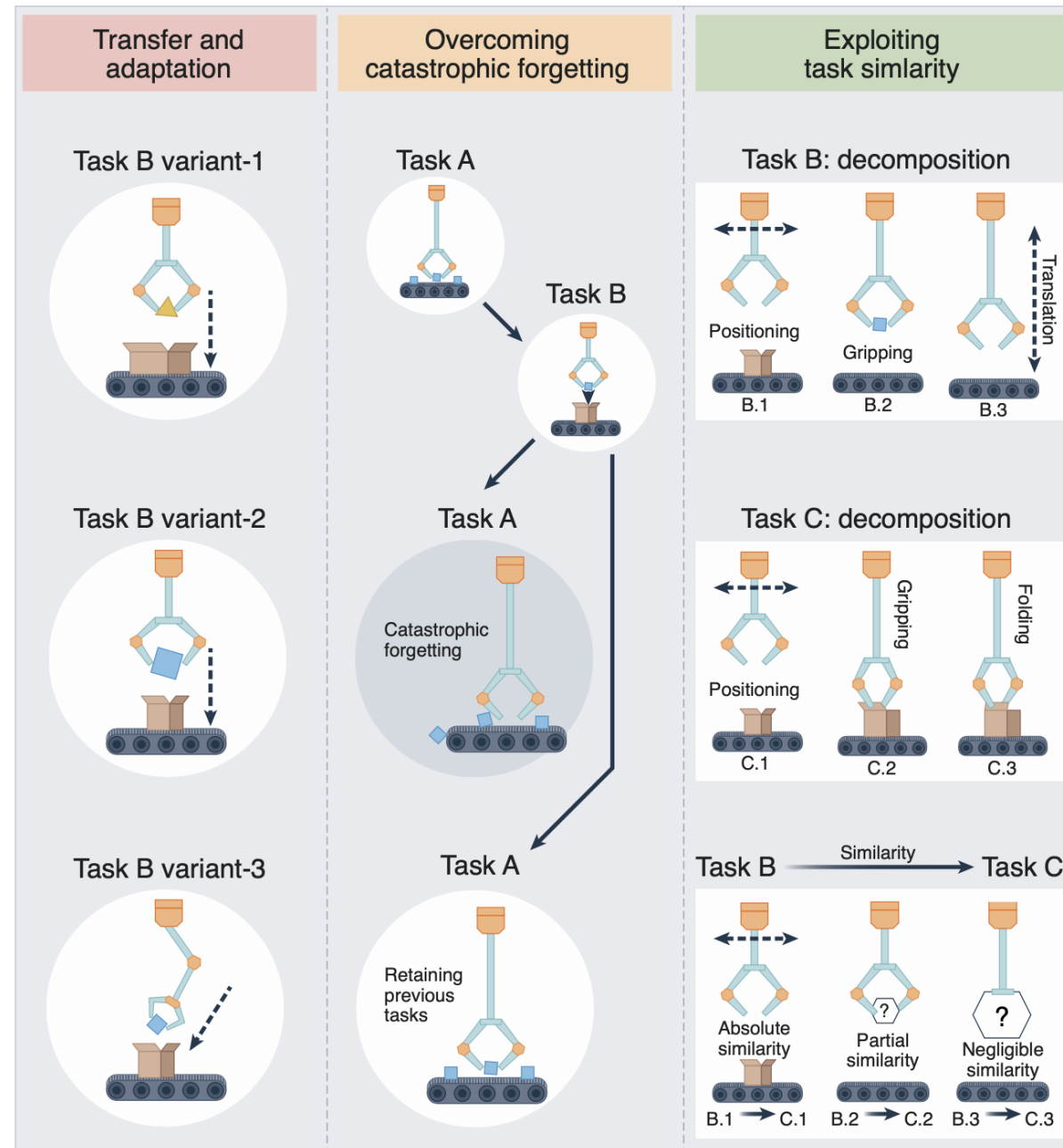


[src: <https://www.nature.com/articles/s42256-022-00452-0>]

Continual learning

Key features

- It is used primarily to avoid catastrophic forgetting while learning from new experiences
- It balances between stability and plasticity, potentially via task similarity





Type of data

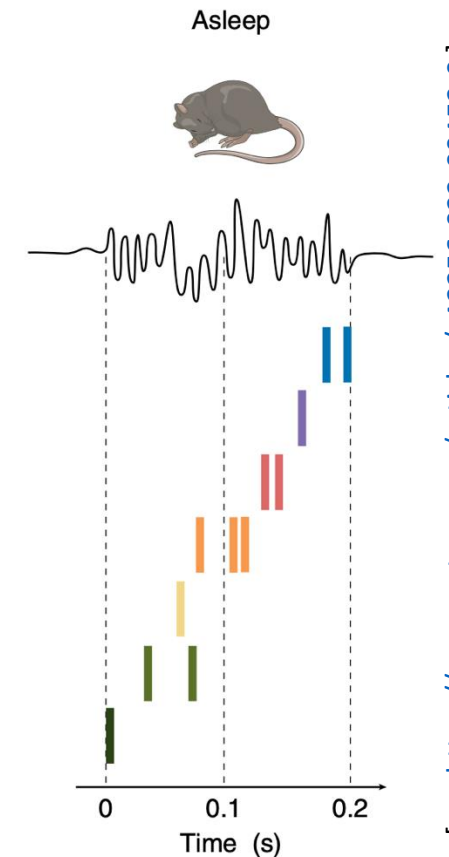
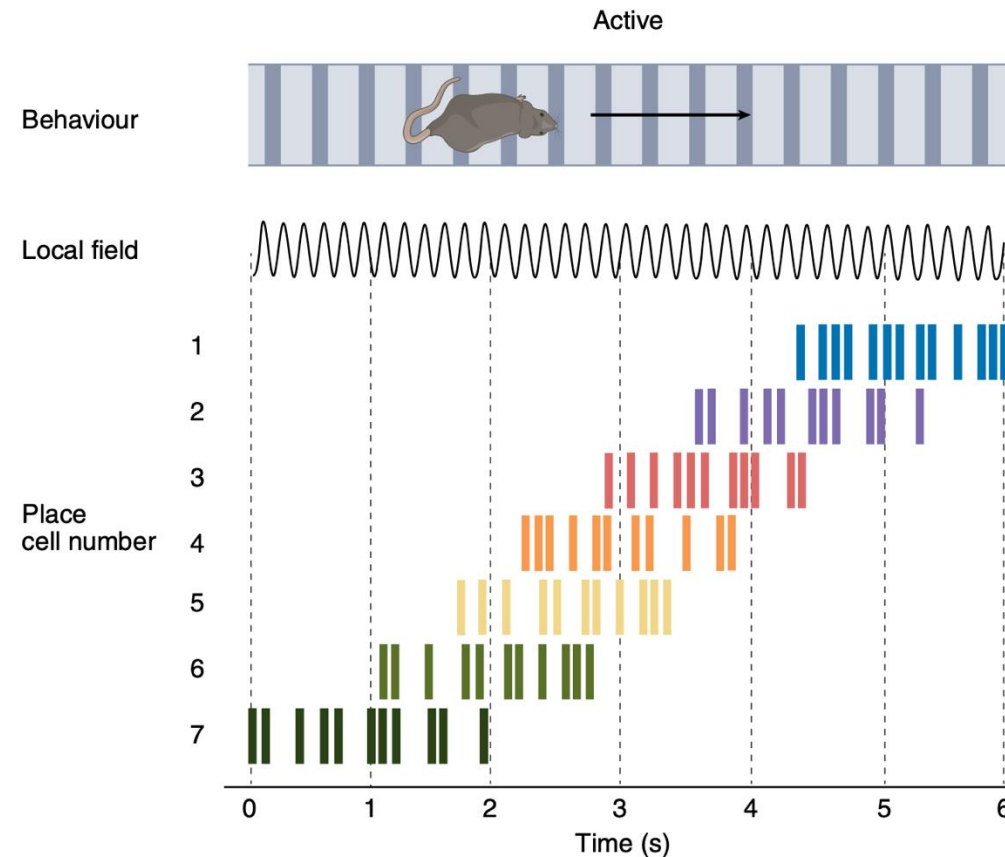
Data arrives in (often manually drafted) **experiences** (sometimes also called **tasks**), where training and test episodes are well separated, assuming **data** points are **i.i.d. only within the same experience**. Data in two experiences are normally distributed differently. Changes are supposed to affect just the input distribution (**virtual/data drift**).

Techniques

Rehearsal strategies (a.k.a., episodic replay)

Alleviate catastrophic forgetting by

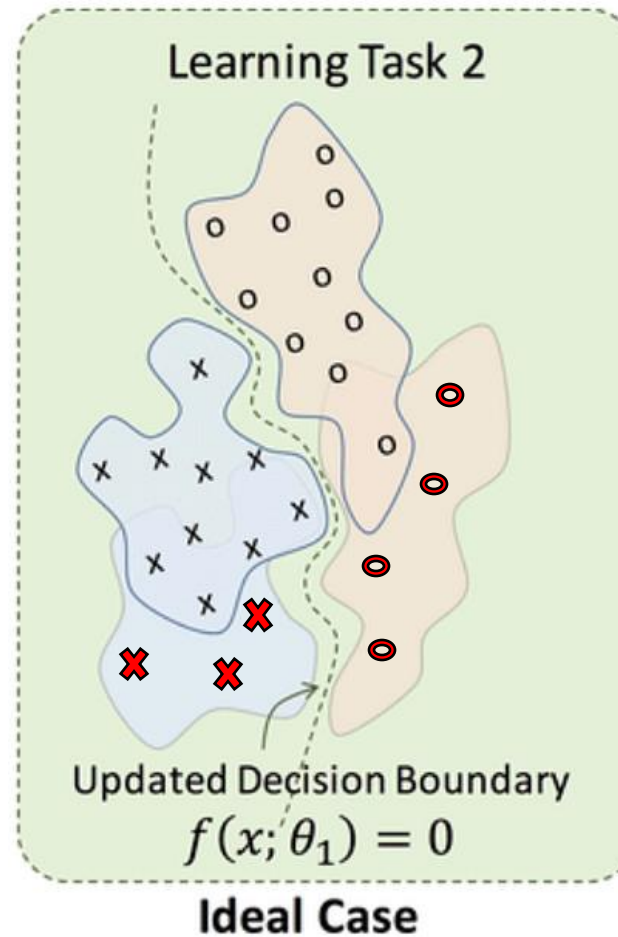
- **storing past data points** in an episodic memory system
- **regularly replaying past data** points with real ones from the new task
- **learning from the** obtained **mix** of replayed and real data points



[src: <https://www.nature.com/articles/s42256-022-00452-0>]

Techniques

Rehearsal strategies (a.k.a., episodic replay)



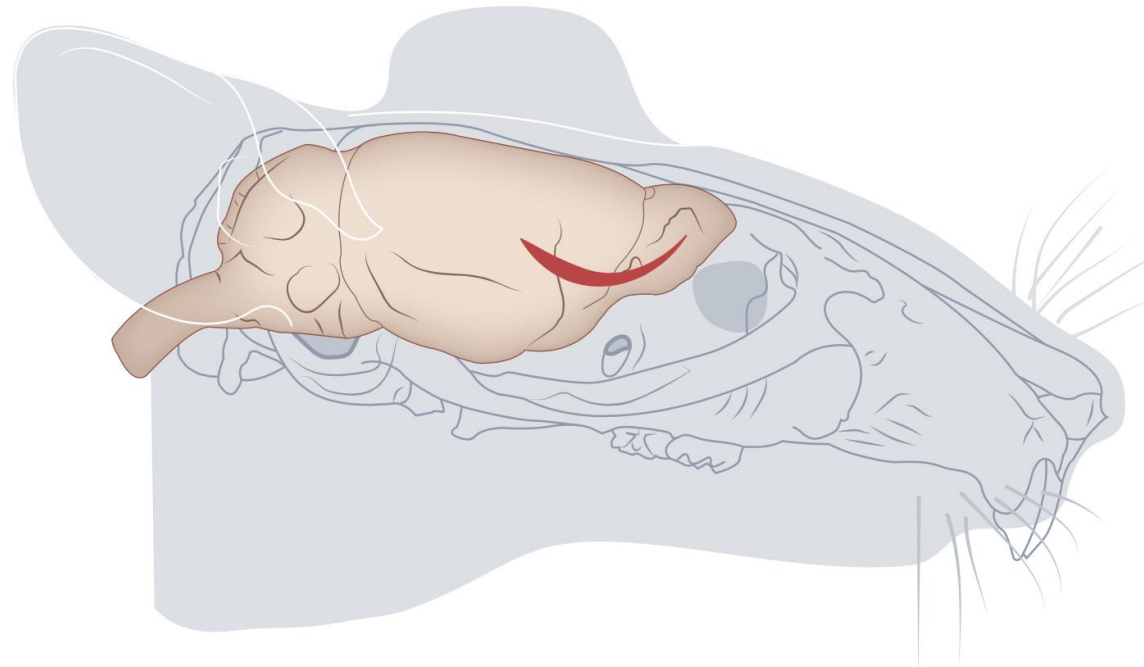
Add to the current task's training set examples from the previous tasks to not forget them

Techniques

Architectural strategies (a.k.a., neurogenesis)

Alleviate catastrophic forgetting by

- choosing specific deep learning architectures, layers, activation functions, and/or
- weight-freezing and **dynamic architectures** (e.g., adding new layers)

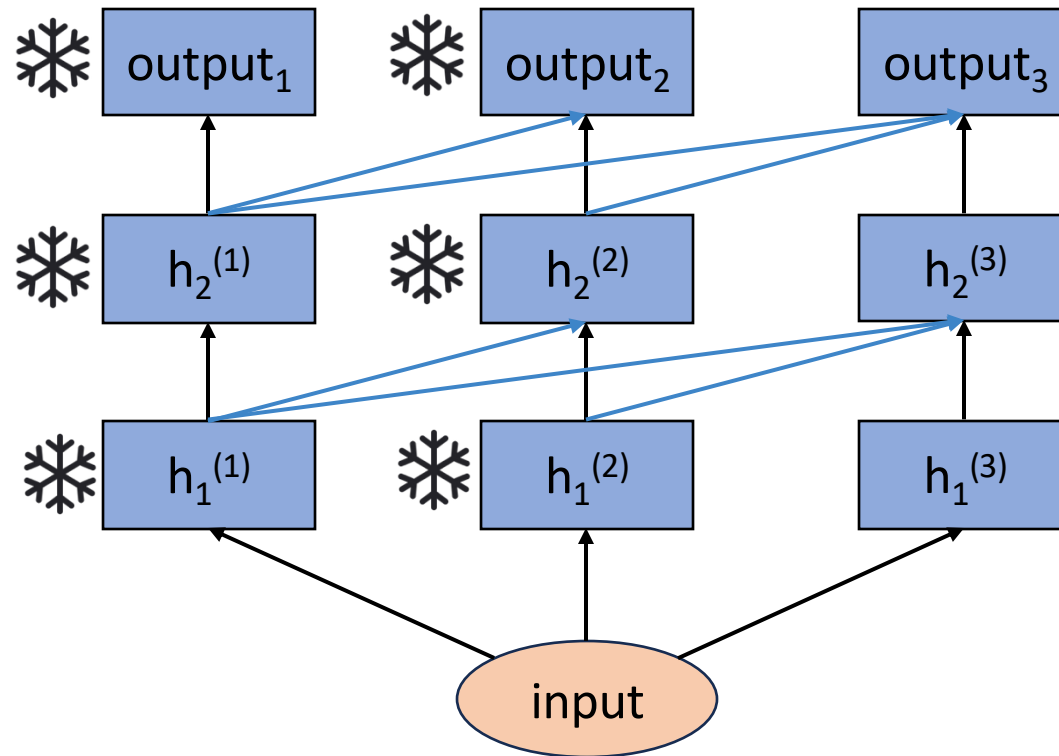


In mice, olfactory interneurons are **produced and** subsequently **migrate** to the olfactory bulb to scale up the number of **new memories** that can be encoded and **stored without catastrophic forgetting** of previously consolidated memories

[src: <https://www.nature.com/articles/s42256-022-00452-0>]

Techniques

Architectural strategies



[src: <https://www.nature.com/articles/s42256-022-00452-0>]

Learning Characteristics

Continual Learning

- ensures that models can **adapt to new information** while **retaining previously learned knowledge**
- **focuses on long-term model stability**



Overall conclusion

In summary, ***streaming machine learning*** is designed for ***real-time data stream processing*** and model adaptation, ***time series analytics*** is primarily ***retrospective*** and focuses on historical data analysis, while ***continual learning*** is concerned with the ***long-term adaptation*** of models to new data while avoiding forgetting essential knowledge.

The **choice** of approach **depends**
on the specific **requirements** of the application **and**
the nature of the **data** being analyzed.

Streaming Data Science
Challenging
the i.i.d. assumption

Emanuele Della Valle

Prof @ PoliMI

CRO & founder @ Motus ml

founder @ Quantia Consulting



POLITECNICO
MILANO 1863